

THE UNIVERSITY OF CHICAGO

THE LANDSCAPE OF DATA SUBJECT ACCESS RIGHTS:
EXPECTATIONS, CORPORATE IMPLEMENTATIONS,
AND NOVEL MECHANISMS FOR USER EXPLORATION

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ABSTRACT

Since the General Data Protection Regulation (GDPR) became enforceable in 2018, dozens of federal and regional governments have passed laws granting millions of people the right to request their personal data from online platforms. Despite how widespread this right has become, the details about how it should be implemented are loosely defined. This dissertation defines how consumers experience invoking this right of access nearly a decade after it has become widespread. I tackled the problem in three studies: (1) a user study to understand consumers' experiences making requests and interacting with their data; (2) a measurement study in which I analyzed and defined a taxonomy for the DSAR mechanisms for 250 online platforms; and (3) a final measurement study in which I collected personal data exports from 69 platforms and analyzed their formats and content.

The findings of the three studies show that every stage of the DSAR, from request to interaction with the data exports, lacked standardization and varied by platform and business type (e.g., travel, finance), as well as across consumers of the same platform. The DSAR mechanisms varied in their complexity (e.g., the number of check-boxes in a form), ease of access and availability (e.g., number of clicks from a privacy policy page to the DSAR), and configurability (e.g., request for data within a date range). In the user studies in this work, I found that lack of clarity and organization leads consumers to be dissatisfied with their personal data exports. While study participants hoped to learn either about their own usage of the platform or how the company collects and uses their personal data, these questions were often left unanswered. I found similar evidence for the barriers to the usability of exports in the export measurement study. In particular, these data exports vary across and within platforms in number of bytes, amount and complexity of file and directory structures, and in the types of records within them.

In addition to describing the intellectual findings from these studies, I release open-source resources and tools for making use of DSARs and data exports in the form of code to process

data exports in a privacy-preserving way, a repository of summarized and generalized data export file schemas from 69 platforms, and library to facilitate data transfers.

Thesis statement: The objective of this thesis is to describe and evaluate how companies implement Data Subject Access Rights mechanisms and to identify key improvements that would make these mechanisms more effective privacy tools for consumers.

CHAPTER 1

INTRODUCTION

For over five decades, legislators have been writing into law a consumer’s right of access and portability. In the former, a consumer has the right to make a request to an entity for all their personal information and, in some cases, how it is being used by that entity. In the latter, the consumer has the right to request that data be ported to a new service either directly or through a format where the user can do this themselves.

While the details of how these rights should be enacted and enforced vary by location, these laws have only increased in scope since their earliest manifestations. In the latter half of the twentieth century, the U.S. Federal Trade Commission signed into law the 1974 Privacy Act which guarantees a consumer the right to access information about them held by federal agencies [1] and six years later the Organization for Economic Co-operation and Development included this right in their fundamental privacy guidelines [62]. More recently, state and federal lawmakers have included it in influential data privacy legislation, such as the E.U.’s General Data Protection Regulation (GDPR) in 2018 [49] and the California Privacy Rights Act (CPRA) in 2020 [89, 90], which grant the same rights to consumers.

While businesses have implemented the ability for consumers to request their personal data for access or portability by submitting Data Subject Access Requests (DSARs) or portability requests [166, 125], there has yet to be a comprehensive evaluation of these implementations. Such an evaluation is vital since work has shown key usability issues in DSAR implementations and usefulness of data exports (also known as data download packages) [173]. In this work, I present three studies that tackle this problem: an evaluation of user interactions with DSAR systems, a large-scale study of how companies implement request mechanisms, and a measurement study describing the data exports consumers receive after making DSARs. In addition, I document the development of the several custom tools I built for these studies.

1.1 Background

The security and privacy community has studied the impacts, implementations, and consequences of data subject rights legislation since the GDPR went into effect. The bulk of this work is done from the implementer’s perspective (i.e., companies complying with laws) and focuses on data protection legal requirements [32, 53, 144, 154]. The inner-workings and techniques through which companies comply with these laws and the challenges they encounter are, of course, important, but not the main focus of my dissertation. Instead, the work that my dissertation builds upon is that concerned with user experiences interacting with this legislation (see Section 1.1) and auditing company practices post-hoc. Also relevant is research conducted in any field that has leveraged data access and portability rights to collect data, as it informed the approach I have taken to collecting consumer data.

Past work has found that consumers have to overcome many obstacles to exercise their privacy preferences and rights. Examples include hunting for opt-out buttons in subscription emails [64, 85], understanding cookie banners [170, 123, 118], or parsing complex choices and jargon in a settings portal [26, 144, 107, 120, 138]. These portals, and more broadly privacy dashboards developed by companies for their users, have been the specific focus of many usability studies. On the one hand, some research has found that they can strengthen trust between a company and its users [52, 8], but they ultimately fail at establishing this trust when it comes to communicating how inferences are made and targeting is conducted [51, 149, 136, 130].

While researchers have conducted similar work applied to data subject access and portability rights in particular, the literature is not quite so extensive. Initial work highlighted vulnerabilities in the DSAR authentication process that can be abused to steal data from others [16, 106, 45, 168, 155, 165]. Researchers have also shown that consumers experience long wait times and feel discouraged to make requests [86, 42, 84]. Additionally, previous work has shown that data exports can be disorganized, though the extent of disorganiza-

tion is unclear [28, 133]. To summarize, the literature suggests that there are many issues, including usability, when it comes to exercising data subject rights. However, the extent or quality of these barriers is not systematically understood.

Due to these quality issues, academics and industry practitioners have explored ways of improving DSARs and data exports. Leschke et al. designed an automated tool for making DSARs, minimizing the burden on consumers [93], though it requires companies to build support for the tool, limiting its potential reach. Others have attempted to streamline a company’s process of responding to DSARs by either providing tooling for data workers [164, 143], attempting to automate the process [176, 112, 82], or studying inefficiencies in GDPR compliance as a whole [151, 152, 104, 145]. Indeed, there are now a handful of companies whose business model is outsourcing the complex work involved in responding to and streamlining compliance with privacy law [72, 74, 75, 121]. Though it has yet to happen in the context of data subject rights, academics and researchers have built custom interfaces for interacting with privacy choices and data tracking when the defaults are unsatisfactory [40, 6, 178, 146]. These works are representative of the first wave of attempts at making the process of exercising data subject access and portability rights less burdensome, but the lack of clarity on how these rights are implemented and the characteristics of DSAR interfaces hinders the ability to build entirely useful tools, which the studies in this dissertation address.

1.2 Chapter Overview

The chapters in this work are organized around the three studies I completed in collaboration with other researchers followed by the technical descriptions of the tools I built for those studies.

Chapter 2 To identify directions for improving the user experience of data exports, we conducted an online study in which 33 participants explored their own data from Amazon, Facebook, Google, Spotify, or Uber. Participants first articulated questions they hoped to answer using the exports. While participants hoped to learn either about their own usage of the platform or how the company collects and uses their personal data, these questions were often left unanswered. After listing these question, participants then annotated parts of the data they found confusing, creepy, interesting, or surprising. Participants’ annotations documented their excitement at finding data records that triggered nostalgia, but also shock about the privacy implications of other data they saw. Having examined their data, many participants hoped to request the company erase some, but not all, of the data. In addition to these findings, I discuss opportunities for future transparency-enhancing tools and enhanced laws.

Chapter 3 In this chapter, I present a large-scale analysis and characterization of the request component of DSARs. I define a taxonomy of the **request** process, developed from an analysis we performed on the interfaces and interactions of 250 platforms. We found that DSAR mechanisms vary in their complexity (e.g., the number of check-boxes in a form), ease of access and availability (e.g., number of clicks from a privacy policy page to the DSAR), and configurability (e.g., request for data within a date range). We find, e.g., that platforms with higher user traffic implemented more complex and configurable DSAR mechanisms.

Chapter 4 This chapter builds upon Chapter 3 by focusing instead on the exports obtained from DSARs. In this study we analyzed 117 exports of personal data files (i.e., **data exports**) of 27 participants, who submitted exports of 69 platforms (a subset of those in the request analysis study, Chapter 3). We found that these data exports vary across and within platforms in number of bytes, amount and complexity of file and directory structures, and in the types records within them. At the extreme, a single export contained over 6,000 files

(X, formerly Twitter), though over a third contained fewer than 5 files. The amount of data in the exports also varied widely, from under one kilobyte to 1.2 GB, though the median was 142 KB. In addition to our findings, I contribute the first repository of anonymized schemas and file and directory structures used in the exports of the 69 platforms from our study to aid researchers, consumers, and legislators in future analysis.

Chapter 5 In this chapter, I describe the technical specifications of the tools I built to accomplish the goals in the studies presented in preceding chapters. I present four tools that consume and process consumers’ data exports and two tools that leverage data portability to enable data transfers (such as data donations). For each tool, I also provide key learnings and opportunities for future research or tools concerning data access and portability.

Chapter 6 In this final chapter of this dissertation I synthesize the work from the preceding chapters and provide directions for future work. I also highlight my most pressing concerns regarding how DSARs are implemented and provide suggestions for how they can be addressed.

1.3 Resources and Tools

Following is a list of the open-source resources and tools I developed as part of the work in this dissertation described in Chapters 4 and 5:

1. Data Export Schema Repository: <https://github.com/UChicagoSUPERgroup/dsar-export-schemas>
2. Pardner: <https://github.com/dtinit/pardner>
3. Pardner-site: <https://github.com/dtinit/pardner-site>

CHAPTER 2

CONSUMERS’ REACTIONS TO EXERCISING THEIR DATA SUBJECT ACCESS RIGHTS

2.1 Introduction

Over the past decade, legislators in many countries have strengthened consumer data protection and privacy rights. Two prominent privacy laws are the E.U.’s General Data Protection Regulation (**GDPR**) [49] and the California Privacy Rights Act (**CPRA**) [158]. Both aim to strengthen consumers’ control over data collection and processing, influencing subsequent legislation in Japan [35], Utah [91], Connecticut [159], and Colorado [61]. The GDPR in particular guarantees four key rights: knowledge, deletion, opt-out, and non-discrimination. In this study, I focus on the right to knowledge, and specifically whether online platforms are respecting this right.

This right to knowledge, or access, is a critical component of U.S. and international privacy regulation and has been for decades, even before the GDPR and the CPRA. The U.S. Privacy Act of 1974 mandates that individuals can “gain access to [their] record or to any information pertaining to [them]” for government-held data [1]. The right to access is one of nine Fair Information Practice Principles (FIPPs) [37] and eight OECD Privacy Principles [62]. This decades-old right is a key mechanism for privacy transparency. By examining their data, individuals can evaluate an organization’s data practices and take action, such as by deleting or modifying data the organization holds about them.

A primary vehicle for guaranteeing the right to knowledge is the Data Subject Access Request (**DSAR**), a communication “addressed to the organization that gives individuals a right to access information about personal data the organization is processing about them” [103, 5]. Once a DSAR is made, consumers receive a **data export**, consisting of one or more files (potentially of many types) containing their personal data. Article 12 of

the GDPR specifies that these data exports must be sent “without undue delay” (at most “within one month of receipt of the request”) and be “concise, transparent, intelligible and easily accessible” [49]. Further, Article 20 requires that these exports be portable, meaning they must be in a “structured, commonly used and machine-readable format.”

Previous work has shown it is challenging to structure a data export in a way that is machine-readable (Article 12), yet still intelligible to consumers (Article 20). Veys et al. conducted focus groups of consumers and showed that companies are not overcoming this challenge [173]. Their participants found data exports unusable or otherwise not useful in their current state. This state of affairs is unfortunate. Prior studies have demonstrated that consumers are interested in, and can gain value from, data exports [173, 177, 8]. Furthermore, tools like Spotify Wrapped, which visualize personal data (not necessarily from data exports), have become popular [142, 156, 163]. Unfortunately, these tools typically visualize only a handful of metrics. For instance, Spotify Wrapped displays a user’s music-listening habits [156], but a data export from Spotify contains far more information. A recent research prototype, TransparencyVis [146], aims to enable general-purpose interactions with an entire data export, but does not appear to have been designed based on users’ goals for data exports.

Thus, I argue now is a good time to (1) determine to what degree exports are adhering to this legislation, (2) determine the degree to which current data exports are meeting consumers’ needs, and (3) pinpoint ways of refining future legislation by identifying what consumers are most interested in learning from their exports. To this end, we conducted a user study in which 33 users of Amazon, Facebook, Google, Spotify, and Uber made DSARs, explored and annotated their data exports using a web app we built, shared pseudonymized versions of their exports, and reflected on their experiences.

In this chapter, I answer the following research questions:

RQ1: How complex or concise are data exports? The GDPR states data exports should

be “concise” and (along with the CPRA in identical language) presented in a “structured, commonly used and machine-readable format.” Concise exports are more accessible for consumers, making it so that they can enact their privacy choices without the burden of extensive searching. To answer this question, we analyzed the 801 data export files 33 participants shared. We found that the complexity of data exports varied both across and within platforms. Amazon data exports typically contained a dozen directories; Spotify data exports contained only one.

Any consumer or data subject should be able to use data exports as a window into a company’s privacy practices. The GDPR codifies this by requiring that the information in data exports be in an “intelligible and easily accessible form, using clear and plain language, in particular for any information addressed specifically to a child.” The CPRA similarly states that data exports should be “easily understandable to the average consumer.” Therefore, we ask:

RQ2: What kinds of questions did participants want to answer with their data exports, and which of these were they eventually able to answer? To investigate, we analyzed questions our 33 participants submitted before exploring their data, alongside their reflections on those questions. Participants were most curious to learn about their usage patterns (e.g., hours spent daily), platforms’ data-collection practices, and how the platforms use personal data. Because of the heavy use of jargon, large exports, and specialized file formats, participants were often left with incomplete or missing answers.

RQ3: How effective are data exports as tools for providing information to consumers? Since data exports are inaccessible to most consumers due to their size, file format, and complexity [173], we built a React-based web application with which participants explored their exports and annotated bits of information they found confusing, creepy, interesting, or surprising. These annotations showed that participants reacted positively to finding records that triggered memories. They also liked finding record types for which they searched (e.g.,

login times). Notably, they looked for a wide variety of records. For example, while some wanted to map out every Uber ride they had ever taken, others wanted to discover new music based on past Spotify listening behavior. In contrast, a number of participants were shocked by the records they saw, feeling their privacy had been violated and their agency undermined. Some participants were especially surprised when they discovered data they thought they had previously deleted.

In addition to the right of access, the GDPR and CPRA give consumers the right to modify incorrect data, erase data, and restrict processing of data for specific purposes (e.g., prevent the sale of personal data to third parties). These rights are how consumers enact their data privacy preferences and are most useful once consumers have the knowledge granted by data access rights. To evaluate their relevance, I ask:

RQ4: Are the actions consumers are most interested in taking after exploring their data in line with those guaranteed by privacy laws? In a post-task survey, participants wanted specific data records erased if they could not justify why the platform would collect that record. In their annotations, participants similarly wondered how they could prevent the platform from collecting analogous data in the future.

Our findings suggest the need for redesigning platforms for exploring data exports to better meet consumer needs. These tools should include a comprehensive list of all data record types in the export. More often than not, basic syntheses and visualizations of each record type will be sufficient. To more closely abide by Article 12 of the GDPR, companies should experiment with different data formats for their data exports and define jargon any time it is used. Finally, data exports could promote user agency and trust by embedding actions consumers can take (e.g., modifying a specific setting or requesting a record be erased) within the exploration process.

2.2 Background and Related Work

2.2.1 *Data Subject Rights Implications for Privacy*

Since the GDPR went into effect, the security and privacy community has been studying the impacts of data subject rights. Many studies have found shortcomings of the GDPR, mostly due to imprecise language, and have developed solutions. Habib et al. found that data deletion options and opt-outs for email communications and targeted advertising are not easy to use [64]. Utz et al. argued that the methods for obtaining consent for cookies should be made more explicit in existing regulation since small implementation decisions (e.g., the positions of cookie banners) have a large impact on user actions [170]. Farke et al. showed that privacy dashboards—interfaces that allow users to review and control data collection that partially parallel the right of access [140]—strengthen trust between users and platforms and are therefore effective tools for communicating security and privacy data practices [52]. Most related to this study, researchers have also shown that consumers experience long wait times [42, 84] and encounter manipulation tactics [86], poorly organized exports, and privacy risks during authentication [28, 167, 155, 165]. Our study similarly asks if the right to access as it currently exists is sufficient to meet consumer needs.

Some studies have focused on identifying adherence to—or violations of—privacy regulations. Degeling et al. analyzed privacy changes on popular websites after the GDPR’s introduction, finding a 16% increase in cookie consent notices on European websites [43]. Bertram et al. analyzed URLs users requested to be removed from Google Search for five years following the passage of a “right to be forgotten,” demonstrating that consumers were enacting their right and Google was complying [11]. Nguyen et al. found that nearly 30% of Android apps sent personal user data to data controllers without the user’s explicit prior consent [116]. In this study, I search for similar instances where platforms could be more compliant with consumers’ right of access.

Academics and industry practitioners have also explored ways of improving DSARs. Leschke et al. designed an automated approach to making DSARs, minimizing the burden on consumers [93]. Others have attempted to streamline a company’s process of responding to DSARs by either providing tooling for data workers [164, 143], attempting to automate the process [176, 112, 82], or studying inefficiencies in GDPR compliance as a whole [151, 150, 104, 145].

2.2.2 Friction in Usable Privacy and Transparency

Past work has shown that explaining privacy practices to consumers and giving them tools to enact their privacy preferences is challenging to do effectively. As it relates to conveying privacy information, previous work has found that consumers are often left confused or dissatisfied after attempting to understand privacy policies and practices [46, 96, 113, 92], in-app algorithmic explanations [114], and advertising mechanisms [26].

When designed poorly, privacy explanations and mechanisms left consumers feeling like they lacked agency over their data and how it was used [66, 107, 87, 181]. Research has shown that consumers were particularly sensitive to the scale at which data collection occurs [140, 148, 160]. This concern was amplified when dealing with more sensitive data, such as location [20] and health data [44, 59, 101]. When consumers felt concerned about how their data is used and powerless to influence it, their trust in platforms decreased, and they report a decrease in willingness to use them [187]. This is particularly important given the growing number of platforms reliant on consumer-generated data to operate (e.g., LLM conversational agents).

On the other hand, when these explanations and tools are written and designed in an accessible way (e.g., by avoiding technical language and lengthy text), consumers are more trusting of platforms [51]. As it relates to the right of access, past work has shown that transparency (e.g., justifying the presence of a specific record in a consumer’s data export)

can also increase consumers’ willingness to stay on that platform [161]. In this work, I highlight areas of particular complexity in the process of making DSARs and as it relates to the data exports. With this information, designers of usable privacy tools can choose to prioritize clarifying and simplifying their data access mechanisms and practices.

2.2.3 Making Sense of Personal Data

Information about a company’s privacy policies and privacy-enhancing features can be hard to find and is often written in complex, inaccessible ways [120, 133, 139]. Previous work has shown that not only do consumers have an interest in enhancing their privacy [109, 48, 33], but tools and interfaces that increase transparency into company data practices lead to increased interest and concern with privacy. For example, Arias-Cabarcos et al. showed that Meta’s “Off-Facebook Activity” Dashboard was effective at informing consumers of previously unknown flows of their data [8]. Datta et al. built a similar interface for Google’s Ad Settings page [40]. Angulo et al. built a tool for consumers to visualize data disclosures, which they similarly found to be clearly informative [6]. Other work has demonstrated the effectiveness of privacy nudges [4], data disclosure dashboards [54, 52], mobile privacy dashboards [130], and data flow dashboards [12]. Most relevant to this study is Schufrin et al.’s TransparencyVis prototype, developed for visualizing data exports [146]. The investigation into users’ goals in this chapter complements their tool and suggests future directions.

2.3 Method

To understand users’ goals and reactions to exploring their data exports, I conducted a two-part online user study. The first part was a screening survey where we asked participants to make DSARs for one or more of the five companies studied (see Section 2.3.1). The main study consisted of participants exploring and annotating their own data exports. The screening survey needed to be separate from the main study for two reasons. First, platforms

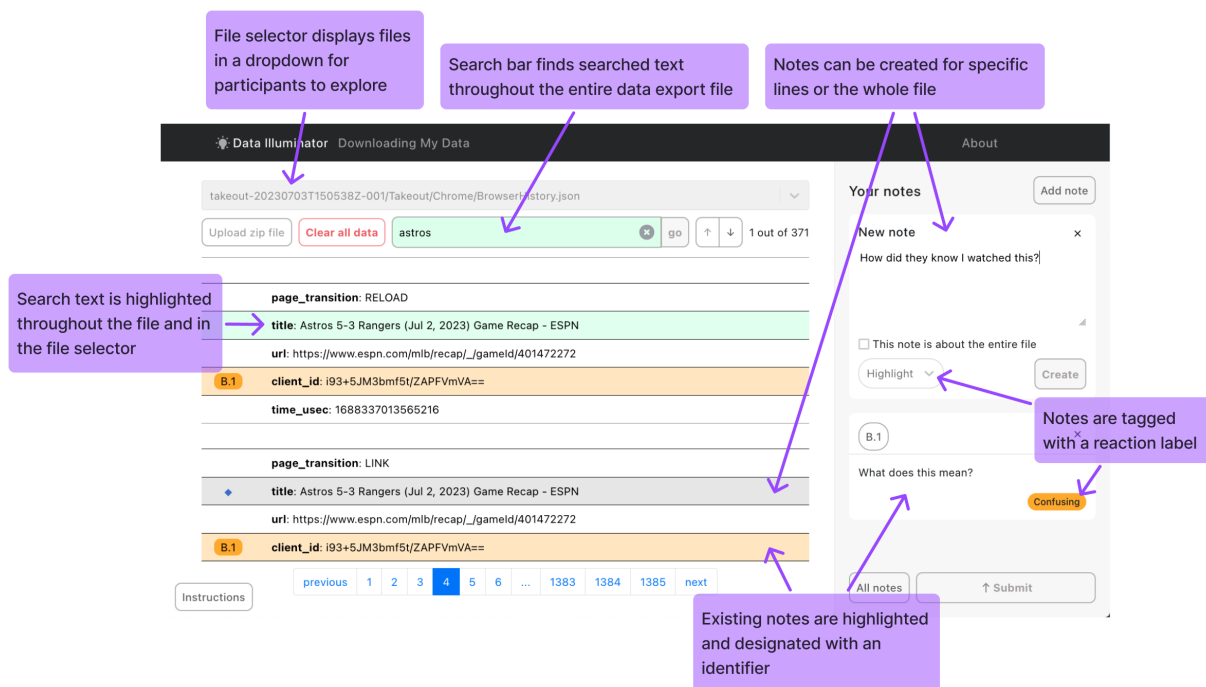


Figure 2.1: Diagram showing the process of interacting with the web-based tool we built for exploring and annotating data exports.

do not respond to DSARs immediately. Second, we wanted to filter for participants who had a significant history of usage with the companies chosen.

In the main study, participants uploaded their data to an interactive web platform we built. We decided to build such a platform because, as previous research has shown [173], most consumers are not familiar with reading or opening the files in formats many platforms use in their data exports (e.g., JSON). Additionally, since we wanted to record detailed and accurate reactions, we wanted to collect these *while* participants were exploring their data. Therefore, our platform included rich annotation features (see Section 2.3.3).

2.3.1 Platform Selection and DSAR Instructions

We recruited users of five platforms for this study: Amazon, Facebook, Google, Spotify, and Uber. We selected these platforms because they are widely used and because they have been a focus of previous work on data exports [173, 146]. Our original study design also included

X (formerly Twitter), but both the research team and pilot testers experienced substantial or indefinite delays in receiving their data exports, so we excluded it.

Because of technical constraints described in Section 2.3.3, our annotation platform could not always accept participants’ entire data exports, which for some companies can include tens of gigabytes of data for active users. Fortunately, except for Uber, the platforms we chose allow consumers to request a specific subset of their personal data. When the option existed, we instructed participants to request their data in JSON or CSV format, as opposed to HTML format. The research team and pilot testers submitted several requests to the five companies throughout the research process to determine the approximate size and types of files contained within these exports. We then used this information to identify categories to exclude from Amazon, Facebook, Google, and Spotify’s data exports. These excluded categories included large, sensitive data like media (photos and videos) and documents stored in Google Drive, as well as the extended streaming history for Spotify. Despite these precautions, some participants’ data exports were still too large for our platform. In these cases, we worked with them to identify folders and files that were requested by accident or unusually large (many gigabytes).

2.3.2 Recruitment and Screening Survey

We recruited participants on the Prolific crowdwork platform. The screening task was open to any English speaker in the world over the age of 18. Recruiting from an international pool meant our participants completed the study under a variety of privacy regulatory regimes. We did so to ensure our eventual recommendations to tool developers and policymakers would benefit a wide range of consumers. Recruiting from multiple countries and languages provided visibility into the structure of data exports from around the world. Table 2.1 shows the characteristics and background of these participants.

This strategy also helped with recruiting more participants for our complex and lengthy

Table 2.1: Participants’ demographics and prior experiences.

Category	n	%	Category	n	%
Gender			Education		
Male	21	63	High school	8	24
Female	9	27	Some coll.	8	24
Non-Binary	3	9	Trade/voc.	2	6
Technical work experience			Bachelor’s	9	27
No	18	55	Master’s	4	12
Yes	15	45	Doctorate	1	3
Country			No answer	1	3
South Africa	10	30	Race		
Poland	5	15	Asian/Pac. Is.	1	3
Italy	3	9	Black/Af. Am.	9	27
USA	2	6	Multi/Biracial	1	3
Portugal	2	6	White	21	64
Greece	1	3	Other	1	3
UK	1	3	Age		
Hungary	1	3	18-24	16	48
Mexico	1	3	25-34	8	24
Czechia	1	3	35-44	5	15
Spain	1	3	45-54	3	9
No answer	5	15	55-64	0	0
Ever downloaded data?			65+	1	3
Yes	12	36			
No	21	64			

study. We directed participants to a Qualtrics survey that asked about their usage habits of the selected companies. Any participant who had been using at least one of these platforms for at least two years was given instructions on submitting a DSAR. We then asked participants to share their experiences requesting their data. All participants were paid \$2 USD for completing the screening survey regardless of whether they qualified for the main study.

2.3.3 Main Study Task

In the main study, we first asked participants in a Qualtrics survey to confirm they received their data export and to answer questions about their previous experiences making DSARs

and exploring data exports. We then asked them to brainstorm five questions they hoped could be answered using their data export. At this point, we asked participants to upload their data export to our web app and begin the annotation task.

We built our data annotation platform using Node.js and React, hosting it on our research group’s servers (for a full technical description of the tool, see Section 5.3.1). Figure 2.1 shows the main interface and features. After they uploaded their data exports, we asked participants to create 10 annotations on the platform (or five if their data export had 20 or fewer files). An **annotation** consists of three elements: zero or more lines within a file, an explanation of why those lines or that file stood out, and one of four tags—**confusing**, **creepy**, **interesting**, or **surprising**—proposed by Veys et al. [173].

Once participants created the minimum number of annotations, they had the option of creating more or beginning the submission process. In the first step in that process, our platform showed the participants their five original questions and asked them to reflect on whether they had been answered. Then, the data annotation platform prompted participants to identify three additional questions they had about themselves, their data, or the company that provided the data export. We paid participants \$15 USD for completing the main study, which took about 35 minutes. We allowed each participant to complete the annotation task for more than one company, compensating them separately for each.

2.3.4 Ensuring Participant Privacy

In addition to receiving approval from the University of Chicago Institutional Review Board (IRB), we took a series of precautions to protect participants’ privacy and anonymity. The design of our study required participants to upload data that, by its nature, contained identifying information. A data export might include emails, full names, addresses, dates of birth, and more. It also likely contains information that participants would consider sensitive and may not want to share, such as their location history or browsing history.

We used two primary strategies to protect participants’ privacy. First, we performed all processing on the client side (i.e., in participants’ browsers). Second, we pseudonymized the data exports before uploading them to our server. As mentioned previously, the research team completed several DSARs for the five companies and found that the keys in JSON data never contained user data—instead they were used for naming types of records. Therefore, we only pseudonymized the corresponding values in the exports by hashing and doing direct character replacement of characters in JSON values and spreadsheet cell entries.

One participant (P20) reached out to us on Prolific with concerns about their data privacy and security. They were worried about exposing financial information to us, our ability to access their data, and whether they might accidentally delete data by making a DSAR. We responded with the same description of the anonymization techniques we put in the tool (see Section 2.7.1). The participant thanked us, said they were satisfied with our precautions, and completed the study.

2.3.5 *Qualitative Data Analysis*

We performed qualitative analysis on participants’ free-text responses to identify themes. Two team members collected responses to the data annotation task and the survey from about 10% of participants and compared themes to develop a preliminary codebook. While this sufficed for identifying primary codes, all four coders subsequently iterated on secondary codes. We then split responses into four sections. One team member served as first coder on all responses, while four separate team members served respectively as second coder. Each pair met and resolved all disagreements. Finally, the first coder performed affinity diagramming on the codes to identify connections between themes. I correct spelling and capitalization mistakes when reporting participant quotes.

2.3.6 *Limitations*

Our study considered only a subset of the files and data that the selected companies have on users. This limitation was due to the technical challenges brought on by relying entirely on client-side computation, a trade-off we believe was important to ensure participant privacy and promote trust. Additionally, we only built capacity for parsing JSON files, CSV files, and (when the file structure was predictable) selected HTML files. All other files were discarded. While this does limit analysis of the size and structure of participants' data exports, most participants did not click through every single file.

Among companies, Spotify and Amazon had the smallest number of participants in our study. We suspect this limitation was caused by these platforms taking a week or two, the longest among the chosen platforms, to respond to DSARs. Participants would sometimes forget they signed up for the study, not notice they received a data export email before the download link expired, or simply turn down the Prolific task.

While we anticipate that our hashing and redaction techniques should prevent most potential privacy leaks, there is no guarantee that identifying information cannot be found in the names of files or in JSON keys even though we never observed this to happen in our own data exports. For this reason, we made ourselves as available as possible to our participants via email and on the Prolific platform. Most participants did not reach out with privacy concerns, but we promptly responded to the one who did (P20); they chose to complete the study.

Additionally, as Table 2.1 shows, our sample is not representative of global consumers and skewed young, white, and male relative to the broader population. Participants may know more about data practices, which could impact reactions to their personal data. In Section 2.3.3 I described how, in the screening survey, we asked participants to articulate five questions they hoped could be answered during the study using their data export. Some participants might not have had any questions about their data and thus generated questions

Table 2.2: A description of the structure and characteristics of participants’ data exports for the five companies we studied.

Metric	Amazon			Facebook			Google			Spotify			Uber		
# participants	2			9			17			3			7		
# unique keys	749			328			1000			56			99		
# exclusive unique keys	650			149			700			3			0		
Per export	min	med	max	min	med	max	min	med	max	min	med	max	min	med	max
# files	27	51.5	76	9	23	56	1	21	53	8	10	10	7	8	10
# unique keys	250	424	598	45	78	188	7	118	545	41	48	55	67	79	99
# directories	5	17	29	4	12	30	2	15	30	1	1	1	4	4	4
Directory depth	-	-	5	-	-	4	-	-	6	-	-	1	-	-	2
Export size (kB)	194	806	1,418	83	976	6,678	5	4,118	38,318	10	260	952	10	1,243	18,966

for the sake of the task. As such, the questions some participants produced may reflect a set of questions they might abstractly hope to have answered somehow, rather than what they realistically expected to be answered by their data exports. Note, however, that the questions generated by the six participants who viewed the data requested for this study before the annotation task were not atypical of those of other participants: they were about general data usage, searching for specific records (e.g., location), and company security practices.

2.4 Results

I first describe our participants and their DSAR experiences (Section 2.4.1) before characterizing their data exports (Section 2.4.2). I then describe the questions participants posed about their data exports (Section 2.4.3) and key themes from participants’ annotations (Section 2.4.4). Finally, I describe participants’ intended future actions (Section 2.4.5).

2.4.1 Participants and Their DSAR Experiences

In total, 33 participants completed the study. Table 2.1 summarizes their demographics. Participants skewed young (24 were younger than 35), male (21), and white (21). The most represented country was South Africa (10), and the most represented continent was Europe (15). Only 14 participants hold a bachelor’s degree or higher, and the majority work outside the technology field. Recall that we permitted participants to complete the data-annotation task for multiple companies; five annotated data exports for two different companies.

Many participants completed a DSAR for the first time for this study. Of the 33 participants, 12 reported that they had downloaded their data from a platform prior to this study (“ever downloaded data?” in Table 2.1). In their written responses, nine participants reported doing so out of curiosity, while three did so out of necessity (e.g., because of work). Although Figure 2.2 shows that most participants did not struggle with the DSAR process, some encountered issues with the delivery timeliness. While we did not formally measure the time to deliver the data export, participants usually had their Google data export by the end of the day. In comparison, Spotify typically took one to two weeks to provide the data.

Google and Facebook, and to a lesser extent Spotify and Amazon, provide controls during the DSAR process, unlike Uber. Facebook users can choose categories of information to include, such as “logged” or “ads” information. Making a DSAR to Facebook or Google requires approximately the same number of steps, but Figure 2.2 shows participants struggled

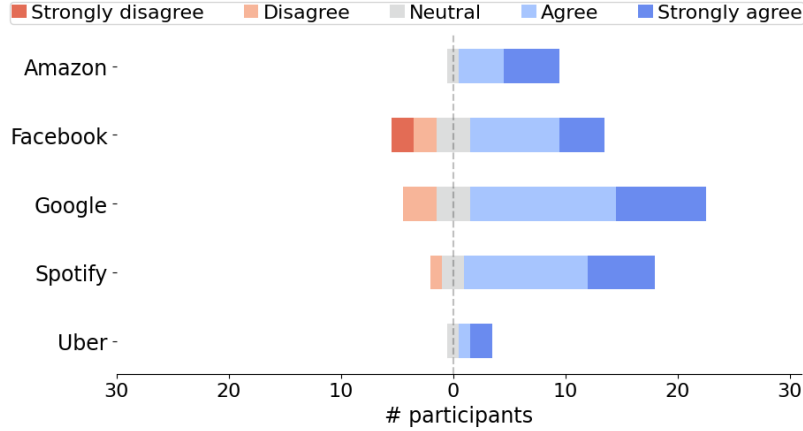


Figure 2.2: Participants’ responses to the following statement: “It was easy to download my data from the platform.”

more with Facebook. This is likely because Facebook’s DSAR instructions were briefly outdated since Facebook changed the DSAR process during the study. We had directed participants to Facebook’s own instructions, and neither of their two pages¹ were updated to reflect the new procedure.

2.4.2 Structure of Data Exports (RQ1)

We received 801 (pseudonymized) files from 33 participants. Table 2.2 characterizes these files. Spotify and Uber’s data exports were the most consistent in the metrics we observed, with two out of the three exports from the former containing files with the same names. The third export was missing the `Identity.json` and `SearchQueries.json` files, although it is unclear why. The identity file in particular contains important data, such as the user’s display name and profile photo.

Google, the platform with the most participants, unsurprisingly showed the greatest variation in data export size and complexity, with a median of 15 directories per participant (“# directories” in Table 2.2). This complexity is understandable since most top-level direc-

1. <https://www.facebook.com/help/212802592074644> and <https://www.facebook.com/help/contact/180237885820953>

Table 2.3: The structure and characteristics of participants’ data exports by region. Five participants did not disclose their location, while “other countries” are the U.S., U.K., and Mexico. The complexity of data exports varied across regions.

Metric	Continental Europe			South Africa			Other countries		
# participants	14			9			3		
# unique keys	1333			480			858		
# exclusive unique keys	872			304			851		
Per participant:	min	med	max	min	med	max	min	med	max
# files	7	26	84	4	11	43	9	31	53
# unique keys	45	78	188	41	48	55	67	79	99
Export size (kB)	10	987	38,319	5	6,724	18,966	858	5,386	7,625

tories are named after individual Google products (e.g., `HomeApp/` and `Maps/`). All Google data exports had a `MyActivity/` directory, which seemed to be a log of different types of activities for different products, although it is sometimes unclear what these activities are. These activity files, along with history files, were among the largest. To our surprise, these history files were not consistently named across Google participants. For instance, the browsing history was named both `Chrome/History.json` and `Chrome/BrowserHistory.json`.

Participants completed our study from 11 different countries (see Table 2.1) and had different language preferences (see Table 2.3 for export complexity across regions). Google was the only platform for which we received filenames in different languages; however, we speculate this is because participants for the other platforms all had set English as their preferred language. Even then, keys and column headers in files with non-English filenames were still in English.

To further understand the complexity of these exports, we counted the number of unique keys within every file in a data export (# unique keys in Table 2.2). The sample file in Figure 5.1, for example, would have four unique keys: `my_best_friends`, `first`, `last`, and `handle`. Amazon and Google’s largest exports both had a similar number of unique keys (598 and 545, respectively), but the median number of keys per Amazon participant (424) was more than double that of Google (118). This means Amazon has a higher density of

unique keys given that the median export size (806 KB) was about one-fifth that of Google’s (4,118 KB). Like other metrics, the number of unique keys varied across platforms.

2.4.3 *Participants’ Goals for Their Data (RQ2)*

Before interacting with their data, we asked participants to think of three to five questions (depending on the size of their exports) about what they wanted to learn from their data. We refer to these as **pre-annotation questions**. After the data annotation exercise (during which participants explored their data, described in detail in Section 2.4.4), we showed participants their pre-annotation questions and asked them to write a few sentences on whether they were able to answer each question; I refer to these as **post-annotation reflections**. While some participants answered with certainty (e.g., “*I did not find the answer in my data*,” P17-Google), others were more vague (e.g., “*there is too much to look into*,” P4-Amazon). We considered pre-annotation questions to be fully answered if the post-annotation reflection contained the answer itself or an affirmative indication that the user found an answer (e.g., “*yes, I got this information*,” P11-Uber). Finally, we asked participants to brainstorm three more questions (**post-annotation questions**) that they still had about their data or the platform after exploring their data export.

In this section, I describe the types of questions participants asked, which of them were answered, and how they evolved with the added context of the data export.

Table 2.4: Key questions participants initially hoped their data download would answer, the number of unique participants (#) who listed that type of question, total questions in that category (?), and the number of questions fully answered (✓) after exploring.

Question Type	#	?	✓	Example Participant Quote
Platform information	29	93	59	
Selling & sharing data	10	17	10	“Does the company share [its] users’ data with other companies?” (P14-Google)
Storage	10	12	10	“How long is my data kept for?” (P33-Google)
Usage (general)	8	10	5	“Why do you gather my data?” (P35-Google)
Business practices	7	15	5	“Does the company use AI to make specialised playlists” (P4-Spotify)
General data questions	7	10	10	“What data about me is being collected” (P6-Google)
Access	7	9	5	“Who has access to my data?” (P33-Google).
Privacy & security (general)	6	9	5	“Is my data safe?” (P16-Google)
User actions available	4	7	6	“Can I ask you to delete my data?” (P15-Google)
Personalization	3	4	3	“Do they use my position data to show me personalized ads?” (P23-Google)
Specific participant records	24	59	15	
Location	11	13	11	“Does Uber track my travel locations?” (P18-Uber)
Interactions with other users	4	5	3	“How many people did I engage with?” (P27-FB)
Nonconsensual data	3	4	3	“Is the company collecting data it said it wouldn’t?” (P22-Spotify)
Devices	3	3	3	“Are the devices where I’ve logged on stored and detailed?” (P3-Uber)
Browsing/search history	3	3	3	“Does Amazon know my browser history when suggesting me stuff?” (P15-Amazon)
Time spent on platform/activities	3	3	2	“Do they keep track of how much time I spend in the app?” (P23-FB)
Payment information	3	3	2	“How are my payment details stored?” (P28-Amazon)
Other records	15	24	16	“How many reports or clarifications have I generated in the app?” (P11-Uber)
Participant information	13	30	7	
Patterns, trends, & hypotheticals	7	9	3	“What locations have I been to based on my location history, and can I identify any significant travel patterns or places of interest?” (P13-Google)
First/most of record	6	9	3	“What was my first friend request I got?” (P12-FB)
Total of record	5	8	0	“How many hours have I spent on Facebook in a certain period of time?” (P27-FB)
Platform’s perception of user	3	3	1	“Compared to other users, am I a good user to the platform?” (P27-FB)

Questions answered by the data export. I first categorize the pre-annotation questions for which participants successfully found answers during the data annotation exercise. Table 2.4 lists the different types of questions along with the number of participants who asked them, the number of questions of that type, and how many of those were answered.

Closed-ended questions about the existence of records in the export and generic questions about the contents of the files were most commonly answered. In total, 24 participants asked 59 questions about whether a specific record type existed in the data (“specific participant records” in Table 2.4). These questions were typically short and simple, often in the form of “*Is my location stored in the data?*” (P28-Amazon), although some used the term “*tracked*” instead. By far the most requested data type was location, with 11 participants (“location” in Table 2.4). Several other participants listed concrete data types like age, browsing history, or payment methods, but other participants were vague when referring to data types.

Data exports were effective at answering these questions about specific records. In fact, most of these questions were answered—only seven were not (two each in “location” and “interactions with other users” and one each in “nonconsensual data,” “time spent on platform/activities,” and “payment information” in Table 2.4). Most participants noticed that the data type in question was being tracked, but their reactions varied. P1, who completed the study for Google and Facebook, found that both companies had “*a bit more*” location data “*than expected.*” P15-Amazon, on the other hand, felt Amazon “*barely*” knew their “*browser history.*”

Participants also asked more general (still often closed-ended) questions about the data. Six participants asked about the security or privacy of their data (“privacy & security (general)” in Table 2.4), such as P25-Google: “*Is my data handled with caution?*” Ten participants asked how data was stored or for how long it was stored (“storage” in Table 2.4), while others asked generic questions (“general data questions” in Table 2.4), like P6-Google: “*What data about me is being collected?*”

Participants who asked generic questions about how much data the platform was storing were able to find answers 10 out of 12 times (“storage” in Table 2.4). Participants found that they were storing “*a lot*” (P6-Google) of data, “*much more than [they] expected*” (P20-FB). In contrast, participants using Uber, Amazon, or Spotify were not quite so surprised, such as P10-Uber, who answered their own question about their Uber trips with “*whole data is here so it’s answered.*” Some answers were quite precise (“*signed in data expires after 26 months,*” P34-Google), others less so (data is kept “*forever,*” P33-Google).

Questions unanswered by the data export. We found that questions that required analysis or computation (“patterns, trends, & hypotheticals,” “first/most of record,” and “total of record” in Table 2.4), questions about a platform’s business practices or technology (“business practices” in Table 2.4), and questions about specific security practices (“selling & sharing data” in Table 2.4) were not answered.

Six participants exploring Uber, Facebook, and Amazon asked about the first or earliest instance of a record (e.g., “*first friend,*” P12-FB, or “*oldest purchase,*” P28-Amazon, “first/most of record” in Table 2.4). These questions were similar to the “specific participant records” since they were short and simple with a concrete answer (in this case, a number), but they did require more analysis or synthesis than simply whether a record was present.

Seven participants asked longer and more complex questions (“patterns, trends, & hypotheticals” in Table 2.4) about their “*communication patterns*” or “*online behavior*” (P13-Google). P24-Uber went a step further: “*Can I use my data to increase the quantity and percentage of discounts I get?*” Participants received fewer answers to these types of questions. No participant who asked about the total amount of money, time spent, or distance traveled over time got an answer (“total of record” in Table 2.4). Additionally, only one of P13-Google’s questions (their “*most frequently visited website*”) was answered. P28-Uber bemoaned the “*lack of summary or any form of analysis,*” which also accounted for the “*disappointing*” lack of answers to P13-Google’s questions about their communication patterns

Table 2.5: After participants explored their data, the questions they still wanted to have answered, including the number of unique participants (#) who listed that type of question and total number of questions in that category (?).

Question Type	#	?	Example Participant Quote
Platform	32	103	
Usage	11	18	“What would they do with information like this?” (P3-Uber)
Privacy & security (general)	9	11	“Is it really safe that they have so much information about the user?” (P11-Uber)
Deleting data & other actions	9	18	“How can I delete all this info?” (P9-FB)
Business practices	9	13	“How is the data used to recommend new music?” (P4-Spotify)
Storage time period	7	8	“How long is all this data stored?” (P7-Google)
Specific participant records	7	9	“Does Spotify collect banking details?” (P4-Spotify)
Confusion	6	9	“What are the strange symbols on some lines?” (P20-FB)
Missing data	5	8	“Why isn’t there a device login history category?” (P3-Uber)
Access	5	5	“Who can access this information besides the user?” (P11-Google)
How do they know?	3	4	“How do they have this data?” (P1-FB)
Others	6	12	
Participant-related	4	6	“When do I use Uber [most] often?” (P26-Uber)
Annotation platform	2	6	“What will you do with my data?” (P33-Google)

and online behaviors. The content of the data exports tended to be mostly raw data—any summaries or syntheses of information were absent. This is likely why most participants were unable to find answers to these questions.

In addition to wondering *what* data platforms stored, participants wanted to know *how* it was used. For example, ten participants wondered if their data was sold to third parties (e.g., “Does Uber sell client data to telemarketers?” P24-Uber, “selling & sharing data” in Table 2.4). Further, 15 questions regarding the platform were about its business dealings (“business practices” in Table 2.4). P4-Spotify, for example, asked, “Does the company use AI to make specialized playlists?”

A little more than half the questions about platforms’ usage of data were answered. There were no patterns where some questions were better answered than others (“usage (general)” and “selling & sharing data” in Table 2.4). In fact, both P25-FB and P9-FB asked if their

Facebook data was being sold for the platform’s benefit. P25-FB concluded “*Yes, by ad partners,*” while P9-FB wrote, “*I don’t know more about this point.*” Out of 15 questions about the business side of the platform (excluding those related to selling or sharing data), 10 remained unanswered, with no indication as to why there was no answer (“business practices” in Table 2.4). GDPR requires platforms to disclose how they use consumer data, but not necessarily within a data export. These unanswered questions could indicate incorrect expectations of what a data export would contain, or a desire to have these exports cover a wider range of topics (e.g., a platform’s privacy policy).

New questions. Compared to pre-annotation questions, post-annotation questions (Table 2.4) focused less on the participant and more on the platform. Only four participants had lingering questions about themselves (“participant-related” in Table 2.5), which were most similar to the “patterns, trends, & hypotheticals” in Table 2.4. In fact, some participants carried questions over, such as P26-Uber, who asked about when they used Uber the most both before and after the annotation task. Only seven participants asked about the existence of specific records post-annotation, compared to 24 pre-annotation (“specific participant records” in both Table 2.4 and Table 2.5), again indicating that data exports in their current form are well-equipped to address these types of questions.

Questions about “usage,” “privacy & security,” “storage,” and “business practices” were consistent between pre-annotation and post-annotation questions. All participants who had these questions before annotation still asked questions in at least one of these categories post-annotation, although the details sometimes changed. Three participants who asked questions about the business operations of the platform became curious about comparing data across companies, such as P3-FB, who asked, “*How does the data in the EU differ from data in the US?*” This pre- and post-annotation consistency indicates that although current data exports may not be suited to answer questions about how platforms use and manage user data, these questions are important to users.

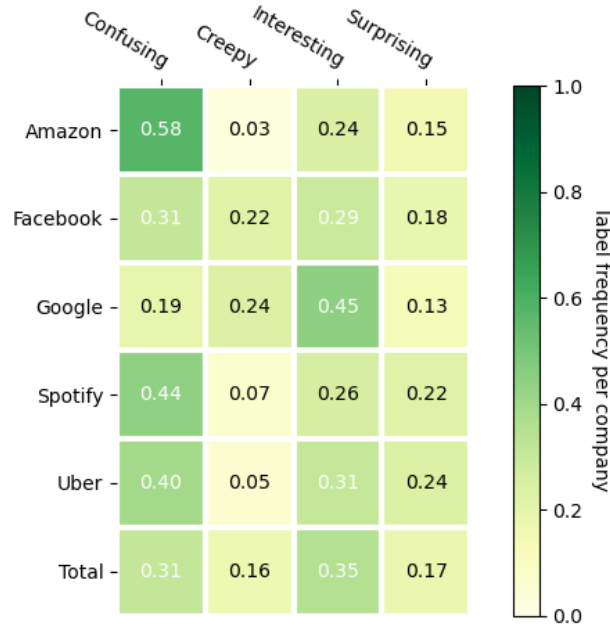


Figure 2.3: The predominance of the four tags by platform. Percentages are with respect to total annotations per platform.

2.4.4 Reactions in Data Annotations (RQ3)

In this section I present findings from the data annotation exercise, the main part of our study. In the exercise, we asked participants to highlight segments of their data that stood out to them, categorize that segment (*confusing*, *creepy*, *interesting*, or *surprising*), and explain why it stood out to them.

Figure 2.3 shows the frequency with which each of the four labels was used. The most common label was *interesting*, making up almost 40% of the labels applied. However, this finding is influenced by the large number of Google exports we analyzed, of which *interesting* was the most common label. For every other company, the most used label was *confusing*, hovering from around 30% (Facebook) to nearly 60% (Amazon) of labels. Participants did not use the *creepy* label often. It only exceeded 20% in the case of Google and Facebook. For all other companies, its use was below 10% of annotations.

Overall, though, there was no drastic difference in the number of times each label was

used. Table 2.6 suggests the reason—participants often had different reactions to discovering similar or identical pieces of information. Consider two participants for the same platform: P1-Google commented that Google collects “*too much info and they store it for too long*” and P16-Google similarly found the “*very extensive information*” in the Google export to be “*a bit disturbing*.” Despite these similar annotations, P1-Google labeled theirs as *creepy* while P16-Google labeled theirs as *interesting*. Thus, I describe high-level themes that crossed label categories.

Surprise the company knows this information. While we never explicitly asked participants about privacy or security concerns before the data annotation task, Table 2.6 shows themes related to these concerns appeared in all four categories. Participants often questioned why platforms had specific records, how they had them, and how long they planned on storing them. Participants sometimes pointed to specific types of records that should not be collected or stored at all.

Most participants (20) discovered types of records that they had not previously known the platform collected (“tracking, collection, & use” across the four labels in Table 2.6). These included spending habits, payment methods, emoji use frequency, search history, browsing history, login location, times when activities started, times when activities ended, IP addresses, phone carriers, devices used, car ownership, and locations where they downloaded apps. Participants responded with surprise and confusion. P6-Uber, for example, was not “*sure why [it] is necessary*” to keep their “*bank details in the payment methods file*.” The level of detail and specificity of the data often alarmed participants, such as P2-FB, who noted “*Emoji count? Why? I just... Why?*” after discovering a file with counts for every emoji they had ever used on Facebook.

Table 2.6: Clustering of participants’ annotations, with the number (#) of unique participants creating each, whether the subject of the annotation is the platform (🏠) or participant (👤), and an example.

Annotation Theme	#	Example Annotation
Confusing	30	
🏠 Missing & incomplete data	14	P16-Google opened their YouTube search history file and noted the history “ <i>is incomplete, I don’t know what is the reason, because I watched more materials.</i> ”
🏠 Value meaning	9	P32-Google found a list of Chrome extension ids and noted “ <i>I don’t understand what is written there.</i> ”
🏠 Key/column naming	8	P2-FB annotated a line in <code>your_record_details.json</code> with “ <i>What the hell is a checkpoint?</i> ”
🏠 Tracking, collection, & use	9	P4-Spotify annotated <code>Inferences.json</code> with “ <i>Looked at all files and I have no clue what they are for. This doesn’t sound like something connected to music at all.</i> ”
🏠 Deleted records still exist	3	P21-Google annotated the <code>Search/MyActivity.json</code> file with “ <i>Google keeps my data from a very long time ago even though I sometimes delete browser history of all time on all my devices.</i> ”
🏠 Inferred data	3	P23-FB found a list of topics Facebook thinks are interesting to them in <code>feed.json</code> and noted “ <i>I don’t know why they think this a interesting topic for me.</i> ”
🏠 Befuddlement	3	P9-FB annotated 4 lines in <code>advertisers_you’ve_interacted_with.json</code> with “ <i>??????</i> ”
Creepy	23	
🏠 Tracking, collection, & use	18	P20-Google has “ <i>no idea why</i> ” the Google Play Store order history would be “ <i>showing locations.</i> ”
🏠 Old/stale data storage	5	P6-Google noted “ <i>Wow that was a long time ago - 6 years of activity tracked?</i> ” when exploring their YouTube activity.
🏠 Preferences violated	6	P25-Amazon noted in <code>consents.json</code> that they “ <i>do not remember consenting to ad partners.</i> ”
🏠 Nonsensical/incorrect data	4	P12-FB “ <i>I don’t even [know] who those people are</i> ” listed in <code>people_who_followed_you.json</code> .
🏠 How to stop collection	3	P9-FB wondered “ <i>how can I delete this?</i> ” in the <code>recognized_devices.json</code> file.
Interesting	33	
🏠 Tracking, collection, & use	13	P25-Google noted “ <i>I did not know that my steps are counted</i> ” in <code>derived_com.google.step_count.delta_com.google(2).json</code> .
🏠 Platform inner-workings	10	P1-FB noted “ <i>I didn’t know this option existed</i> ” in <code>ad_preferences.json</code> .
👤 Self-discovery	9	“ <i>It helps me visualize when I started on Uber Eats. It has been longer than I thought</i> ” (P18-Uber in <code>eats_restaurants-0.csv</code>).
👤 Memory & recognition	7	“ <i>It reminds of the beautiful pictures I captured back then</i> ” (P27-FB, <code>eats_restaurants-0.csv</code>).
🏠 Against data collection	8	P3-FB noted a file with administrative records “ <i>seems excessive to store.</i> ”
🏠 Missing & incomplete data	8	“ <i>Entries are less detailed than anticipated</i> ” (P18-Uber, <code>eats_restaurants-0.csv</code>).

Continued on next page

Table 2.6 continued

Annotation Theme	#	Example Annotation
 Confusion	6	P13-Google noted in the video searches activity file that “ <i>This could have the real title to be more easy to identify.</i> ”
 Useful data	5	“ <i>Detailed information about used phones in your account. Very useful information</i> ” (P16-Google, <code>Devices.json</code>).
Surprising	29	
 Tracking, collection, & use	14	“ <i>They have my address and I don’t like it</i> ” (P1-Google, <code>Addressesandmore.json</code>).
 Missing & incomplete data	7	P11-Uber noted that “ <i>In such a complete report, they should have</i> ” IP addresses in <code>rider_app_analytics-0.csv</code> .
 Nonsensical/incorrect data	6	“ <i>I’m not sure I downloaded these apps</i> ” (P1-Google, <code>GooglePlayStore/Library.json</code>).
 Self-discovery	5	“ <i>I listen to quite a lot of rap</i> ” (P19-Google, playlist video file).
 Memory & recognition	3	“ <i>This is one of the games I played with some friends about 6 years ago</i> ” (P12-FB, <code>instant_games.json</code>).

Some participants felt like their preferences had been violated, noting they either opted out of or never knowingly opted into data collection. P20-FB, for example, wondered, “*So does the location pick me up even though my location is not activated???*” (`where_you’re_logged_in.json`). P21-Google, plus three other Google participants, realized “*deleting browser history does not mean that [their] data is removed forever.*” P2-FB also felt their preferences were violated when they encountered their off-Facebook activity file:

“*Words can not really express. Why is Facebook tracking my off Facebook activity? How? Why? Where did I agree to this? How do I opt out? This is by far the most disturbing thing.*”

Sources of confusion and barriers to comprehension. Participants were sometimes unable to interpret the meaning of components within a data export, including file names, keys, and values. Nine participants were confused by the meaning of the values in their data exports (“value meaning” in Table 2.6). Some of these values were identifiers (e.g., `orderId`, `advertiserName`) and dates (e.g., P13-Google said their `birthday` key was set to 0000), but in most cases our pseudonymization prevented us from investigating further (e.g., `description:*****.....@`).

Eight participants could not decipher the meaning of some keys within their data exports (“key/column naming” in Table 2.6). P23-FB, for example, did not “*understand what they mean by ‘off Facebook activity’*” in `your_off-facebook_activity.json`. Other keys that caused confusion were `UsedForDisbursements` (in `payment_methods-0.csv`, Uber), `PrimeStudentMarkedForGraduation` (in `Subscriptions.PrimeTransition.csv`, Amazon), and `voting_location` (in `voting_location.json`, Facebook). No participant mentioned finding the meaning of these terms in their data export.

Some participants (14) did not find data they were expecting (“missing & incomplete data” in Table 2.6). Many keys had blank or null values for no clear reason (e.g., P8-Spotify said `episode` was set to null in `Playlist1.json`). Participants could not understand why some records had “*such careful tracking of events*” (P18-Uber) and nothing for data participants expected the platforms to have (such as P18-Uber’s order history for Uber Eats). Two participants mentioned seeing incomplete Spotify listening history. Here, the missing data was expected as we asked participants to exclude streaming history from Spotify DSARs (see Section 2.3.1).

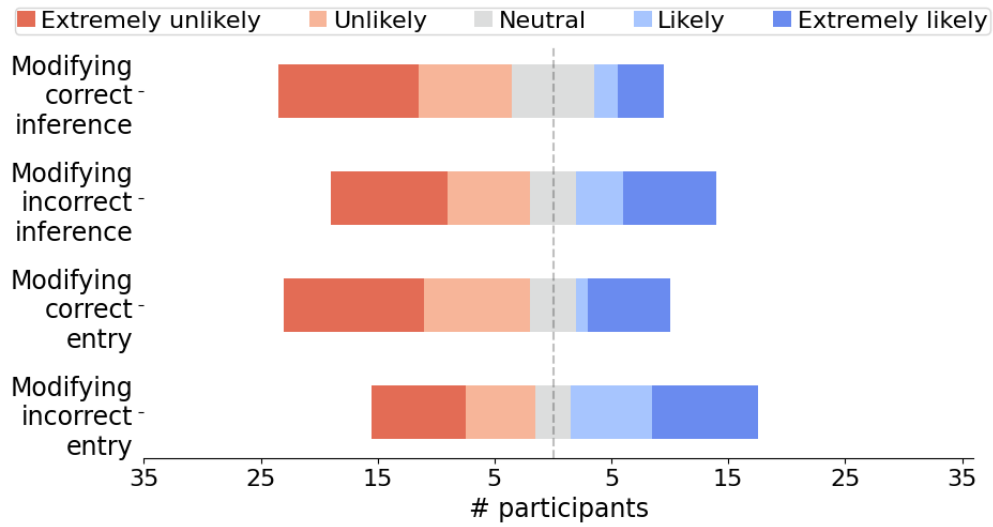
Sometimes when data was present, participants noted it was wrong or nonsensical (“non-sensical/incorrect data” in Table 2.6). Participants noticed some locations listed in their data had no relation to them. P13-Google, for example, noted they “*don’t even know*” the address listed under their saved locations, and asked, “*How can it be my home address?*” In other cases, participants could not remember taking actions recorded in the exports. P23-FB, for example, did “*not remember these people nor saving any products on Facebook*” in the `your_saved_items.json` file. Many of these notes were framed as questions, indicating participants are lacking key information to fully understand the data in their exports.

Self-discovery, memories, and nostalgia. Some data exports helped participants learn about themselves or reflect on important memories. For example, 10 participants enjoyed making discoveries about the advertisers and interest categories associated with them

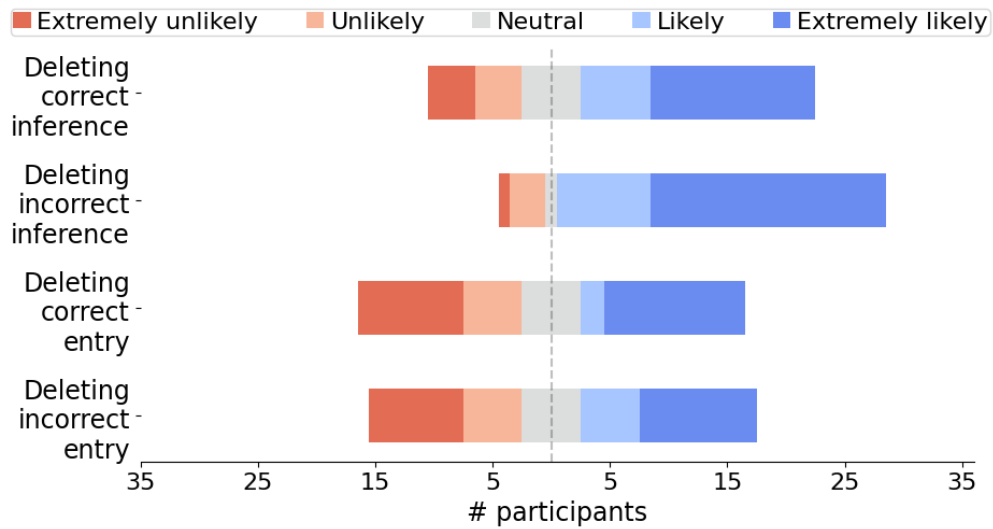
(“platform inner-workings” in Table 2.6). P28-Amazon “*had no idea these many brands bought audiences with [them]. Some of these [they]’ve never heard of*” while exploring `Advertising.AdvertiserAudiences.csv`. Sometimes these findings were amusing, like for P22-Spotify, who noted “*lol, I guess I’m not very profitable. I like that*” on the inferences (here, likely age group) the platform had made about them.

In other cases, participants learned about their own behaviors and attitudes (“self-discovery” in the “interesting” and “surprising” categories in Table 2.6). For example, P19-Google noticed they “*watch quite a lot of educational content*” and “*listen to a lot of rap*” on YouTube. Analyzing data longitudinally, P11-Uber was “*surprised to know the waiting times and [idle times] have increased over the years*” when exploring their Uber trips data. These examples show participants found useful—or at least interesting—information.

In addition to learning new information, eight participants recalled specific events or people after seeing a related record in their data (“memory & recognition” in the “interesting” and “surprising” categories in Table 2.6). While some of these annotations were neutral (“*I remember visiting [this page] on my phone,*” P33-Google), others had emotion. P27-FB, for example, “*loved*” the Facebook story they found in `stories/story_reactions.json`. P12-FB found a “*childhood friend of [theirs] which [they] put as brother in FB ’cause [they] were very close*” while looking through `profile_information.json`. Not all of P12-FB’s memories were positive ones: they later found people “*deleted from their friend list*” whose “*name reminds [them]*” why they were removed.



(a) Likelihood to **modify** data.



(b) Likelihood to **delete** data.

Figure 2.4: Participants' reported likelihood to modify or delete data based on correctness and whether it was either inferred by the platform or entered by the user.

2.4.5 Deleting Data and Other Actions (RQ4)

In their pre-annotation questions, annotations, post-annotation reflections, and post-annotation questions, participants mentioned wanting to take action. The most commonly requested action was deleting some, not all, of their data. While only two pre-annotation questions about data deletion were left unanswered, sometimes answers from separate participants contradicted each other. Two Google participants reached separate conclusions when wondering if consumers could delete their data—one noted it “*doesn’t seem*” like a possibility (P13-Google), while the other answered “*absolutely*” (P34-Google). If there were instructions for making data deletion requests in P13-Google’s data export, they could not find them. Some participants, such as P2-FB, expressed skepticism about whether platforms would follow through:

“If they do delete, how much of this will be actually deleted and not just kept anonymously?”

Participants sometimes thought data deletion was something platforms should implement automatically. P3-FB thought their contacts “*should be wiped after some time maybe*” and P1-Google thought Chrome’s browser history was stored “*for too long.*” P21-Google suggested a solution:

“I think there should be a way to remove my history across all Google platforms by just deleting on the Chrome browser. Also, there should be a choice by myself as a user as to which data about me I want kept by Google, thus being in control of my data.”

Participants were also concerned with future data collection and usage. For example P9-FB asked, “*How can I withdraw my consent?*” from the apps connected to Facebook listed in `connected_apps_and_websites.json`.

Our findings from the post-annotation survey confirm that participants were interested in data deletion. In the survey, we asked participants how likely they would be to modify or delete the following: an outdated home address they entered into the platform, an accurate home address they entered, an incorrect political party inference by the platform, and a correct political party inference. Figure 2.4 shows that participants were more interested in data deletion than modification. Figure 2.4b shows that participants were less likely to delete data they entered themselves compared to data inferred by the platform. Some said their home address was “*shared public information*” (P18-Uber) and therefore not a privacy violation. Others believed the information was relevant to the platform (e.g., “*this data seems to be important for Uber,*” P10-Uber), but this attitude was often company specific—meaning they may be content with Uber having their information, but “*Spotify shouldn’t have [their] address*” (P38-Spotify).

2.5 Discussion

In this study, we asked participants to annotate and answer questions about data exports resulting from DSARs to five different companies (Amazon, Facebook, Google, Spotify, or Uber). We then analyzed their survey responses, annotations, and the exports themselves in order to characterize the content and variability of these exports, as well as how participants wanted to use them and which content they found galvanizing.

Overall, participants were surprised to find specific data (e.g., search history) being collected or retained by platforms. They mentioned different facets of tracking found in several files within the exports. Participants were able to use their data exports to answer simple questions, but also had many more complex questions that data exports are not currently equipped to answer. Participants also expressed interest in integrating actions (e.g., deleting or correcting data) directly into the workflow of viewing their data exports.

Next, I discuss the implications of our findings as they relate to the GDPR [49] and

other laws with similar provisions. I then provide suggestions for developing new tools for consumers to explore data exports.

2.5.1 Policy Shortcomings and Recommendations

Reduce time to deliver. Article 12 of the GDPR requires that platforms respond to DSARs “without undue delay” (30 days) [49]. While all five of the platforms we investigated responded to participants within 30 days of the DSAR, the delay in these responses varied between hours (Google), hours to days (Facebook, Uber), and days to weeks (Spotify, Amazon). After delays of days or weeks, some participants missed the email notification that the data export was available. When they found it, the link had expired, requiring these participants to send another DSAR. **Requiring companies to reduce the time consumers have to wait for DSAR responses to at most 15 days would likely improve accessibility.**

Provide data summaries and definitions. In Section 2.4.2, I showed that many data exports are not concise, an explicit requirement of Article 12 of the GDPR (data must be “concise, transparent, intelligible and easily accessible”). Some exports are large (in Google’s case, sometimes over 30 MB worth of text-based files) and have several labels for different records, demonstrated by the dozens to hundreds of unique keys in JSON files. The data is not always transparent or intelligible; participants were often confused by JSON keys (e.g., `HorizontalAccuracy`) and their associated values and did not understand the meaning of specific records, and sometimes even entire files. **Companies should be required to provide definitions for every file, term, record, and key in the consumer’s preferred language, as a significant step in transparency and intelligibility of data exports.**

I recommend policymakers incorporate the “data dictionary”—a system for providing

information (e.g., a variable’s possible values, human-readable names, variable definitions) about variables in a dataset [27, 56]—into legislation to standardize how this information is communicated. Striking a balance between concision and clarity is more complex since platforms are obliged to send consumers all the requested data, even if it spans many gigabytes and hundreds of files. Therefore, in addition to the entire data export, **platforms should be required to create record summaries**, which could include the total number of records of each type or the earliest date a specific record can be found in the export.

Justify data retention. The GDPR, under Article 15, also guarantees that data subjects receive “confirmation as to whether or not personal data concerning him or her are being processed” and if so, the “purposes of the processing” [49]. Table 2.5, however, shows that participants’ primary remaining questions after exploring their data exports were precisely about how their data was being used. Our results suggest these questions are both important to consumers and difficult to answer with current data exports. Along with providing definitions for records, **platforms should be required to provide a clear description of how the records are used alongside the records themselves.**

2.5.2 Designing a Data-Exploration Tool

We expect platforms and tools built for personal data exploration to continue growing in popularity [142, 156, 163]. Based on participants’ reactions, I suggest that future designers and developers focus on the following three areas. While our data tool (Section 2.3.3) was a means to an end by helping participants explore and annotate their data (and therefore not the focus of our study), I also reflect on how our tool could be adapted to embody these three principles.

Data visualization and interaction without truncation. Participants’ questions often involved discovering patterns in their usage of the platforms. P28-Uber, for example, asked:

“Can you show a summary of my data or perhaps graphs and analysis to understand how I use Uber?” I suggest future tools do just that—provide ways of interacting and visualizing personal data, which many existing tools already do. However, many participants were confused and disappointed by missing data in their exports. While this is likely due to the DSARs they submitted excluding specific categories of data, it shows that participants want every piece of data to be accessible to them. Developers of these tools will need to come up with creative ways of showing aggregated data while also letting participants delve into specific records in the export.

While our tool did not truncate any records within the files it parsed, we did not provide any visualizations or data syntheses. As a result, the experience was overwhelming for participants. In an early prototype of the tool, we abbreviated some of the data by limiting lists within JSON objects to three items and CSV tables to ten items. While this was effective at reducing files that sometimes were split into thousands of pages to fewer than ten, the pilot participants often mentioned wanting access to all of their data. I recommend developers explore the trade-offs between abbreviation and comprehensiveness, but ultimately not sacrifice the latter.

Designing for meaningful interaction. Synthesizing and summarizing data to show users the broader picture is important, but it is not the only way our participants wanted to interact with their data. Our participants had strong emotional reactions to individual records that triggered memories and recognition. P19-Google, for example, was able to find a record for an important milestone: the “*moment when [they] started listening to audiobooks not in [their] native language.*” Other individual data points triggered surprise or alarm by showing participants that their data was collected or used in unexpected, sometimes upsetting ways. Advanced exploration tools could add features designed to surface these highly salient individual events or records (positive or negative), perhaps by taking advantage of advances in machine learning to answer complex questions or identify likely candidate

events. Doing so could enable data exports to provide not just simple transparency, but also understanding [21].

Our tool did not save participants’ annotations on the client side once the study was finalized, but future tools could implement bookmarking for participants to retain records of interest. While we did not ask if participants would be interested in such a feature, allowing them to save records might be a straightforward, but meaningful, interaction.

Enabling action in-band. Participants expressed strong interest in taking action in response to things they found in their data exports, including deleting existing data and withdrawing their consent for a platform to collect similar data in the future. Such a feature was entirely absent in our tool as its purpose was exploratory. The right to delete personal data (“erasure,” Article 17) and to withdraw consent (Article 7) are both granted by the GDPR [49]. Currently, consumers who want to take these actions after reviewing their data exports must visit the platform and find potentially difficult-to-access options and settings. A better interaction mode would enable consumers to take actions directly, in-band, as soon as they decide the actions are needed. Tools should build in mechanisms for taking these actions. Of course, this will depend on the ability for independent data-exploration tools to communicate these desired actions on behalf of users via an API or other automated submission, as has been done for the right to access [93]. It is unclear whether platforms have incentives to support this feature absent further regulation.

2.6 Conclusion

Recent legislation in several countries has made it easier for consumers to access their personal data collected by companies. However, prior work has shown that consumers are often unsure of how to make sense of the data they receive, are overwhelmed by the amount, and sometimes cannot even open the data files [173]. In order to develop better ways for

consumers to interface with their personal data, either directly through the files they receive from a company or through a tool built for users to interact with their data with more ease, we set out to understand what participants want to learn from their data and what their primary reactions are when exploring it. To answer these questions we designed a study where consumers of five different platforms (Amazon, Facebook, Google, Spotify, and Uber) interacted with their data exports firsthand to share their questions and reactions to seeing their data. Part of this study involved the development of a data annotation tool where participants were able to apply labels (confusing, creepy, interesting, or surprising) and attach notes to segments of their data exports that stood out to them.

Participants were primarily interested in learning about their usage of the platform, what data the platform collects, and how the platform uses that data. While most participants were able to find the types of data the platform collects in their data exports, questions about the platforms' use of the data and participants' usage of the platform were often left unanswered after exploring the data exports. Participants had a wide range of reactions to exploring their data. For some, fond memories were triggered when they encountered a record. Other times, participants were surprised to learn about the scale and detail of the personal data held by the platform. As a result, participants were interested in learning how to make requests to have parts of their data deleted.

Based on our findings, I suggest that companies improve the clarity of their data exports by providing easily accessible definitions for files and terms. For designers of tools for personal data exploration, I provide a characterization of different types of questions participants wanted answered as a source for building relevant data experiences for consumers.

2.7 Additional Materials

2.7.1 *Text Excerpts from Annotation Tool*

The annotation tool included the following text explaining the techniques we used to ensure participant privacy:

When you submit your files, we don't receive any of the values of your raw data. For each file, we receive two anonymized versions: one anonymized through hashing and one through direct character replacement.

Hashing replaces every value with a randomized string that allows us to see how often a value appears in your files but not the value itself. For example, if "USA" were hashed to "89ac13af" we would be able to see how many times "89ac13af" appeared in your files but not know that it corresponds to "USA."

Structural anonymization replaces every alphabetical character (A-Z or a-z) with * and every number with \#. The email address `abc@xyz.com` would become `***@***.***` and the phone number `555-123-4567` would become `\#\#\#-\#\#\#-\#\#\#\#`.

All of your data is stored on your browser until you hit the submit button, and anonymization happens before your files are sent to us. If you have any questions, please reach out to [*contact address*] or message us on Prolific. All set? Submit below!

The pre-study task reads: "Before you started the study, you proposed the following 5 questions you'd want answered by now. For each of them, reflect on the extent to which you have an answer after exploring your data and how you feel about the answer or lack thereof."

CHAPTER 3

DESCRIBING THE DATA SUBJECT ACCESS REQUEST

3.1 Introduction

As I describe in Chapter 1, thousands of companies have implemented data access mechanisms to consumers for nearly a decade since the enactment of the GDPR [49]. While businesses have implemented the ability for consumers to request their personal data [166, 125], previous work has found that the request process and personal data consumers get back is not the most easily accessible [19, 173]. Moreover, it is not known how similar implementations of this right are across companies and across regions with differing legislation. In this chapter, I present a study that measures and describes how 250 platforms worldwide have implemented the mechanisms behind making access requests.

While details about a **Data Subject Access Request** (DSAR) process vary by jurisdiction, the GDPR was the first large-scale law to guarantee this right and, therefore, is a strong reference point for many other pieces of legislation that have come since [81]. While I primarily discuss the GDPR in this chapter, note that U.S. state laws (e.g., CPRA [89, 90]) and other international laws (e.g., Brazil’s LGPD [147]) stipulate similar requirements. According to the GDPR, DSARs are made by a consumer, or the **data subject**, to an organization, or the **data controller**. If the controller meets certain criteria, it must collect and identify the usage of the data subject’s **personal information** and provide the data subject this information, which I call a **data export** (also commonly known as a data download package).

Submit Data Request

Select the data that you'd like to be included in your data package below. You can learn more about each of these selections [here](#).

Note that some selections, like Messages, may delay the delivery of your package, and it may take us up to 30 days to collect your data. We will send you an email to the address you registered with when the package is ready.

☐ Account
☐ Your Activity
☐ Activities
☐ Ads
☐ Messages
☐ Servers
☐ Support Tickets

I've changed my mind

Request My Data

(a) An example of a **dedicated form** mechanism from Discord, a social networking, technology company [71]. Accessible from any screen post consumer authentication.

Contact Us

[Preamble]

Your Name *

Your Email Address *

Subject *

Your Message *

[CAPTCHA]

SUBMIT

(b) An example of a **generic form** mechanism from the United Nations [115]. Preamble and CAPTCHA omitted for brevity.

- website visitors can manage their privacy preferences by visiting [Manage Privacy Settings](#)
- customers can manage their privacy preferences in their [Account Settings](#) page, and
- you can always reach us at privacy@godaddy.com or by mail at the addresses listed in [Contact Us](#) to exercise your privacy rights.

(c) An example of a **manual contact** mechanism from GoDaddy's privacy policy that instructs consumers to reach out via email or mail to make a DSAR [73].

Figure 3.1: Variation in DSAR process exemplified by three mechanisms for making the request of varying complexity.

Even with the guardrails provided by legislation like the GDPR, many terms are left undefined (e.g., “accessible form”) and procedures are vague (e.g., through what mechanism should data controllers send a data export?). This often results in different interfaces and processes, as we show in Figure 3.1. Past work has explored details around the implementation of other privacy rights [166, 118, 114, 31], yet only a subset focuses on the right of access. Existing prior studies identified onerous aspects of the request experience common not just to platforms for which consumers are primary users (e.g., social media [19, 173]), but also people search websites with a different target user base [162].

I ask two research questions related to the DSAR request experience: (1) What is the process for making a data subject access request? (2) What features are common in these request processes and how do they impact usability? We tackled these in the study presented in this chapter by recording and analyzing DSAR procedures for 250 platforms [100, 88]. Our findings from this study produced the following contributions: **(1) a list of attributes present in DSAR mechanism implementations** and **(2) an analysis of these attributes on 250 platforms**.

3.2 Background and Related Work

3.2.1 *Translating Privacy Policies into Practice: No Single Approach*

Implementing privacy policies and legislation has always been a challenge and past work has shown that this can impact the quality of these implementations. The literature has focused especially on how compliant websites have been with web-tracking and cookie-consent policies and has found that there is still a significant gap in compliance [166, 68, 105, 123, 154]. Cohen et al. found that absolute deletion has complex implications for machine learning models that have already been trained on that data [32]. They proposed leveraging differential privacy mechanisms to guarantee near inability to distinguish between outputs of models

where the data subject is present and those where the data subject is absent. Ferreira et al. developed a system for guaranteeing data protection requirements that can easily be added to existing web applications [53]. There are many parallel tools, by researchers and industry, to accomplish the task of complying with privacy legislation. This may result in a lack of standardized approaches to the same goal, as I show in this chapter.

Indeed, companies are required to comply with an increasing number of regulations since hundreds of countries and regions within countries have their own privacy legislation [122, 55]. In 2025 alone, nine pieces of legislation related to data privacy were passed in the United States alone [24], occasionally requiring specialized implementations. For example, CCPA restrictions [89], unlike those of the GDPR and the U.K. GDPR [49, 126], are additionally scoped to for-profit entities [102]. These laws are also frequently amended and sometimes lack concrete details that would aid in implementation [90, 127, 50], resulting in non-compliant systems [134, 98].

In our work, we expand the current understanding of how organizations implement privacy laws by describing how they implement the right of access, in particular.

3.2.2 How DSARs Fail to Meet Consumer Needs

As early as 2020, less than two years after the GDPR went into effect in Europe, work in a variety of fields, including computer science, began to show the gaps in implementation and usefulness of the right to access [5, 133, 79]. In an early study conducted with German platforms, researchers found that about a fourth of the platforms to which they made DSARs did not respond [84]. Other studies that followed focused on different aspects of the request and export process. In 2023, Pöhn, et al. created user accounts on 27 different platforms and measured response-time latency and deceptive patterns in the information and mechanism necessary to make DSARs [132]. Take, et al. conducted a similar study, while limiting themselves to people-search websites, but including evidence that data deletion requests are

often poorly implemented [162].

This work builds upon these studies in two key ways. First, I conduct an analysis of DSARs on a much larger scale than any study before. This helped us achieve breadth and reliability in our data. Additionally, I define and measure the different features and components of DSAR mechanisms by building a comprehensive taxonomy, which can be used to evaluate mechanisms beyond those covered in this study.

3.3 Methods

In this section, I describe how we selected platforms for the study (Section 3.3.1) and how we collected and analyzed data from the request side of making a DSAR (Section 3.3.2).

3.3.1 Platform Selection and Filtering

We used the same list of platforms for collecting data on DSAR processes and data export formats. To compile this list, we began from the top 5,000 websites and apps from Tranco [88]. We then performed automated GET requests to those domains to filter out non-consumer-facing websites (e.g., `trbcdn.net`, ranked 87 on Tranco at the time). Later, we combined the remaining Tranco domains with the top 150 overall apps and top 150 trending apps from the Apple App Store and Google Play (through SimilarWeb [100], a web intelligence market-research tool). We selected platforms from the latter two because we wanted newer and up-and-coming platforms to be represented. From those, we identified 61 duplicate platforms (e.g., Google Docs and Google Slides have the same DSAR mechanism as Google), five platforms with burdensome request processes (e.g., confirming identity by providing driver’s license or other documents), and 104 platforms without DSAR mechanisms or whose websites were inaccessible. We ran this study on the remaining 250 platforms.

Table 3.1: The features and characteristics of DSAR forms that were measured for the platforms in this study.

Measure name	Definition
Date customization	
Date range options	The dates for which you can request your data.
Complexity	
Data category toggles	Any selectable item that includes/excludes a category/group/type of data.
Selectable elements	The number of options given for anything in the form. E.g., a drop-down with three options has three selectable elements.
Time	
ETA	Platforms provided estimate for when the request should be completed.
ETA warnings	Similar to “ETA,” but only if the ETA is shown as a warning when an element or option is selected (e.g., a wider date range).
Expiration	If the export (or export link) expires.
Data format	
Has format options	If the data format is at all toggle-able for products or categories.
Format options	All the possible file/data formats.
Accessibility	
# clicks from settings	Minimum number of clicks needed to get to the DSAR page from the main settings page.
# clicks from privacy policy	Minimum number of clicks needed to get to the DSAR page from the main privacy policy page.
Data category definitions	If any of the data categories in the “Data category toggles” have explanations attached.
Delivery	
Destination options	List of the places data exports can be sent to (e.g., email).
Recurring requests	The option for making recurring requests.
Size options	List of the possible sizes the export can be sent in.
Authentication	
Requires password re-entry	Asks the consumer to reenter their password before making a request
Has email confirmation	Asks the consumer to check their email to confirm the request
Requires sign in	If the consumer needs to be signed in to make a request.
Has country selector	If the consumer is required to enter the country in which they live.
Has state/region selector	If the consumer is required to enter the state or region in which they live.
Restrictions	
Has location restrictions	If there are restrictions to where the consumer must live to make a request and, if so, what those restrictions are.

3.3.2 DSAR Data Collection and Analysis

Three researchers began the process of making a DSAR for all 250 platforms. As I discuss in Section 3.4.1, there were three mechanisms for making DSARs: through a dedicated form (designed for data subject requests), through a generic form, or via manual contact to a representative of the platform (each exemplified in Figure 3.1). Two researchers went through the motions of making DSARs for every platform, but only finalized the request if (1) they were an existing consumer of the platform or (2) if not an existing consumer, then the platform must have a dedicated form. I discuss our decision-making in Section 3.3.5. These researchers took screen recordings and captures of the processes of making requests for platforms with a built-in DSAR mechanism, which they found by searching from both the privacy policy and settings landing page. For platforms without dedicated DSAR forms, we recorded the contact information that platforms provided in their privacy policy for privacy matters or exercising data subject rights (e.g., the Data Protection Officer’s contact information).

Codebook development. We iteratively developed our codebook of DSAR measures (see Table 3.1) by engaging in several rounds of exploratory open coding. First, two researchers randomly selected four platforms to develop the initial codebook: two from the top 10 and two bottom 10 forms from the 157 dedicated forms in the data set. After taking screenshots of every page and feature of each mechanism, the researchers then independently listed the features and components of the forms. The two researchers then combined these two lists of measures after discussing them together, resolving initial disagreements on definitions, scope, and name. Secondary codes were created when the coders agreed more granularity would be desirable. E.g., initially we only recorded whether each mechanism let consumers choose file formats, but later added another code to actually record those file formats. When this happened, the modified codebook was extensively discussed between coders and every

platform was re-coded with the modified codebook.

Applying codes and resolving disagreements. Each DSAR mechanism was independently coded by two coders using the codebook described above. The coder’s first step was finding the privacy policy page and the main settings page (if it existed) for the platform. For each starting point, they then recorded their screen as they searched for the location where they could complete or find instructions for completing the DSAR (when coding manual contact mechanisms). Once they arrived at the mechanism they also took a screen recording of the DSAR, expanding each drop-down menu and selecting every option available to get a comprehensive view of the mechanism. Using the screen recording, they applied codes from the codebook.

After a mechanism was independently recorded and coded by two coders, every disagreement was resolved over the course of several live meetings. When necessary, coders compared screen recordings and only moved on from the disagreement when an agreement was reached. Because every disagreement was discussed and resolved, we did not compute a measure of disagreement.

3.3.3 User Study

Part of the data collection for the study described in the next chapter (Chapter 4) involved study participants making DSARs for a variety of platforms they used. We recruited these participants from the crowdwork platform Prolific and required that they speak English, be at least 18 years old, use at least one of the platforms we pre-selected for the study for at least one year, and have over 10 completed tasks with an acceptance rate over 95%. Participants also had to live in a place with active legislation granting them the right to access their personal data.

In one survey, we gave instructions for participants to make DSARs on platforms they

Table 3.2: A categorization of the 247 reasons why participants found it easy or hard to make a DSAR.

Reason category	Count	Explanation
Ease factors	207	
Low effort	97	The mechanism required few steps or the steps were easy.
Useful instructions	82	The instructions we provided as part of the study were helpful.
Low time commitment	13	The DSAR steps did not take long.
Discoverability	15	Finding the DSAR mechanism was easy.
Challenge factors	40	
High effort	24	DSAR required too much writing, too many steps, overly complex steps, or broadly too much effort.
High time commitment	6	The DSAR steps took too long.
Poor communication	5	Long wait time for a confirmation email or never receiving one.
Technical difficulties	5	Experienced an issue filling out the form that prevented the request.

used and asked them about their experiences doing so. Specifically, we asked each participant to rate the ease of their experience making each request via a Likert-scale question and to justify their answer in an open-ended question. To analyze the open-ended responses, we engaged in many rounds of exploratory, open coding to build and refine a codebook. Before analyzing the data, two coders defined two high-level themes: (1) aspects that facilitated the DSAR process and (2) aspects that made the DSAR process more challenging. Then, one coder randomly sorted responses and inductively defined sub-themes for each high-level theme until saturation was reached.

This coder then presented the codebook to a secondary coder and theme definitions were refined together. The second coder then independently coded the subset of codes used to develop the codebook. These initial disagreements were then discussed and resolved and no modifications to the codebook were done. The remainder of the responses were coded by the coders independently and disagreements were, once again, discussed and resolved. These reasons are shown in Table 3.2.

See Section 4.3.1 for a full description of the user study procedure, participant selection, compensation, ethics, and limitations.

3.3.4 *Limitations*

Data collection for the study was hindered by the state in which the researchers conducting data collection resided in, since it does not have legislation granting the right of access. As we describe in Section 3.4.2, some platforms restrict the ability for residents not covered by this requirement to make requests, as was the case with some platforms to which researchers tried to make DSARs (e.g., Walmart and Target). This decreased the pool of exports for analysis and inclusion in the repository.

Finally, it is important to highlight that the analysis on requests and exports are representative of the moment in time in which that data was collected. We note, for example, that while collecting data on the Major League Baseball platform in mid-2025, consumers had 13 forms to choose from.

3.3.5 *Ethical Considerations*

In this section, I report on the stakeholder-based ethics analysis I conducted to evaluate the study in this chapter. I do so by listing the different stakeholders we identified and, for each one, articulating the harms we considered, the mitigations we developed against these harms, and how our procedure and decision to conduct the studies ultimately upheld the four principles in The Menlo Report [169].

The primary subjects of our research are the tools and systems built by companies (operators of the platforms in our study) to comply with privacy legislation that requires them to enable Data Subject Access Requests (DSARs).

Harm: undue burden of workers at the platforms. In 2021, a privacy research group began sending automated emails to public websites with questions about their compliance with privacy legislation [108]. The owners of some of these websites interpreted the emails as security risks or legal threats. As a result, the study was halted and eventually terminated.

We noticed our procedures could result in similar misunderstandings and misused resources by the platforms, so we took the following mitigation. The researchers involved in this study only submitted DSARs for platforms with dedicated forms under the assumption that the generation of the resulting data export would be automated rather than manual. We did this with the explicit purpose of minimizing harm to the platforms.

Harm: platforms’ reputations could be negatively impacted by our findings.

Given a core aspect of our research was to analyze how platforms implement DSAR mechanisms for their platforms, we considered the possibility that our findings might indicate the approaches taken by some platforms may not be the most consumer-friendly. We concluded that platforms that responded to DSARs would, at the minimum, be compliant with privacy laws, therefore our critique of their implementation would not put them in jeopardy of being singled out by enforcers of those laws. Platforms that did not respond, however, might be. Therefore, our mitigation for this harm was: I do not publish the names of platforms that did not respond to the automated DSARs we made. While publishing the remaining findings may hurt the image of some platforms to some consumers, we concluded that the benefit to understanding how DSAR processes could be improved outweighed that potential harm.

3.4 Characterizing the Data Request

In this section, I describe how the process of making data requests varied across platforms by measuring the complexity and accessibility of their DSARs. We did so by analyzing 266 DSAR mechanisms from 250 platforms (some platforms had more than one mechanism for different locations; see Section 3.4.2). I compare these requests along axes like their popularity (via Tranco [88]), business sector (e.g., technology), and measures we defined and collected for this study. I also present the codebook we developed, which defines the characteristics and features of DSAR mechanisms (see Table 3.1).

Table 3.3: The distribution of the three DSAR mechanisms across the categories of platforms in our study. Shaded cells show that only two categories had minority dedicated forms.

Category	Total		Dedicated		Manual		Generic	
	#	%	#	%	#	%	#	%
Tech.	64	25	42	65	21	32	1	1
Games	38	15	13	34	24	63	1	2
Business	31	12	20	64	11	35		
Lifestyle	31	12	20	64	11	35		
News	22	8	15	68	6	27	1	4
Entert.	19	7	13	68	6	31		
Shopping	15	6	14	93	1	6		
Reference	14	5	5	35	8	57	1	7
Travel	10	4	9	90	1	10		
Finance	6	2	6	100				

3.4.1 The Three Ways of Making a DSAR

I define three categories of DSAR mechanisms: dedicated forms, generic forms, and manual contact. A **dedicated form** makes explicit mention of data subject rights in its title, description, or elements. A dedicated form can be quite simple and minimal. For a Bit.ly (link-management technology company) consumer, for example, logging into their account then clicking a “Request report” button in their settings page is all it took to make a DSAR. On the other hand, dedicated forms may also have text fields, menu options, or checkboxes, as is the case with Discord (Figure 3.1a). Some platforms even gave names to their DSAR systems (e.g., Google Takeout and Duolingo Drive-Thru/Data Vault). While some dedicated forms gave consumers personalized and configurable options, they were often long, complex, confusing, and inconsistent.

The other two DSAR mechanisms, **generic form** and **manual contact**, rely on simpler mechanisms, such as email and general purpose forms. A **generic form** is a general-purpose customer service or contact form that platforms also want consumers to use to make a DSAR. The United Nations’ privacy policy, for example, instructed consumers with “questions or queries regarding this data protection and privacy policy” to use a contact form that makes no mention of data subject requests (Figure 3.1b). The most rudimentary mechanism (**manual**

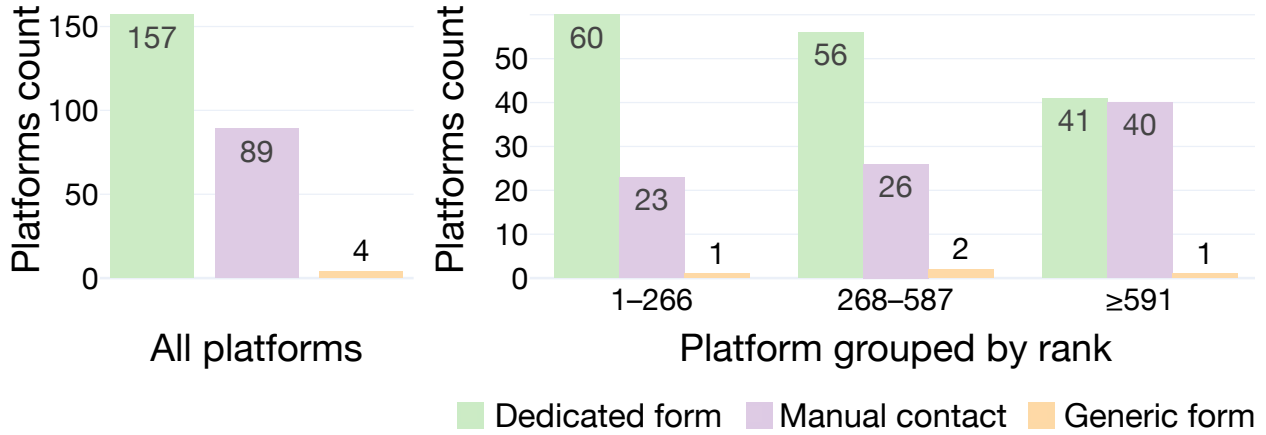


Figure 3.2: The number of platforms (grouped by Tranco rank) that implement each of the three different types of DSAR processes: dedicated form, manual contact, and generic form.

contact) provided consumers with a contact (such as the data protection officer’s email) to whom consumers made the DSAR (e.g., GoDaddy processed DSARs based on requests to “privacy@godaddy.com,” Figure 3.1c). In both the generic form and manual contact mechanisms, a consumer must generate a written request, for which online services [117] and templates [39] exist. These mechanisms can be powerful for consumers sending out mass requests as the content of the request varies little across platforms. On the other hand, they require multiple rounds of communication with a human (e.g., data protection officer) via email, mail, or phone.

Nearly two-thirds (63%) of the DSAR mechanisms we analyzed were dedicated forms, followed by manual contact (36%), and generic forms (< 2%). We found that more popular platforms were more likely to have built dedicated forms, possibly due to more resources and a larger consumer base. Figure 3.2 shows that, among the top third of platforms, over 70% of mechanisms were dedicated forms, compared to 66% in the middle third, and just over 50% for the remaining, lower-ranked platforms. Among 10 business sector categories, only Game and Reference platforms had a minority of dedicated forms; in the remaining categories, it made up 64% (Business and Lifestyle) to 100% (Finance) of the platforms (Table 3.3).

Some platforms used third party services to aid in processing DSARs, which may have

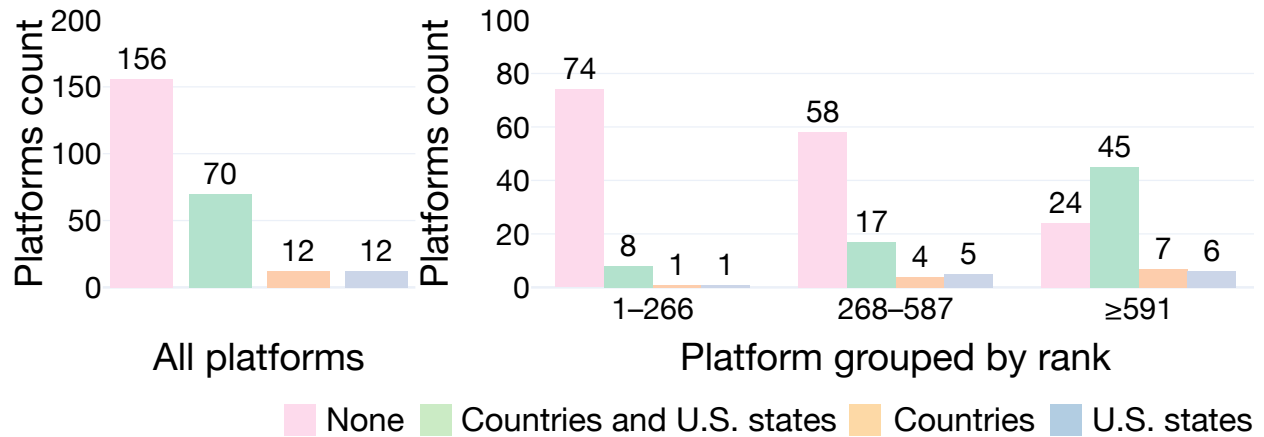


Figure 3.3: The platforms (grouped by Tranco rank) with requests restricted to countries, U.S. states, both, or neither.

dictated the entire dedicated DSAR form flow. We considered platforms with dedicated forms to be using such a service if at any point during the request process an email or webpage mentioned another company (e.g., “Powered by OneTrust” in Taco Bell emails) or if any of the links led to a third-party domain([https://privacy-central.security.ai/\[...\]](https://privacy-central.security.ai/[...]) from Southwest Airlines). It is not possible to know whether a platform uses such a vendor behind the scenes, so we categorized all other platforms as unknown. Figure 3.6b shows that nearly one third (32%) of platforms with dedicated forms used a third party vendor. OneTrust was the most represented among vendors (at 71%). Notably, a OneTrust dedicated form is clearly identifiable by its consistent forms: a single page with identical form-element styling (see Figure 3.4), CAPTCHA, and, as I discuss in Section 3.4.2, identity verification mechanisms. This standardization may help consumers who have previously used a OneTrust form, but any problems in a standardized form will be replicated across platforms.

*** I am a (an)**

Applicant	Customer	Partner
Employee	Former Employee	Other

*** Select request type**

Access my data	Erase my data	Rectify/correct my data
Object to processing my data	Other	

Example of 6 "selectable items"

Mention of data access means this is a "dedicated form"

(a) This form has 11 selectable elements and is a dedicated form since it mentions data access directly. <https://privacyportal-cdn.onetrust.com/dsarwebform/86c88b66-1c6e-4a7d-8530-6d754439e01a/8e724022-6d62-40e3-8de9-2c682113a567.html>

*** Is this for a California Resident?**

Yes	No
-----	----

*** The California resident is or was an**

Applicant for an F5 job	Employee / Former Employee of F5	Employee of a Customer
Employee of a Partner	Other	

*** Select the right you wish to exercise:**

Access My Information	Delete My Information	Correct My Information
-----------------------	-----------------------	------------------------

Location restricted to California

(b) This is a location-restricted dedicated form. <https://privacyportal-cdn.onetrust.com/dsarwebform/86c88b66-1c6e-4a7d-8530-6d754439e01a/ced68d4f-d055-45d3-9fba-27b2d949b5b6.html>

Figure 3.4: Variations in DSAR form for the same platform for different locations.

3.4.2 *Location-Restricted Requests*

Not all DSAR mechanisms were available to all consumers. About 20% of platforms in our study limited requests to consumers who reside in countries or U.S. states with specific, cited privacy laws (Figure 3.3). For example, Starz is a U.S.-based television streaming service that only operates within the U.S. It is unsurprising, then, that its dedicated DSAR form required consumers to select one of 19 U.S. states (those with privacy legislation) as their residence before making the request [157]. For other platforms, the reason for location restrictions was less clear. Wikimedia’s (Wikipedia’s parent organization) privacy policy [179] singled out residents “located in the European Economic Area” and in the United Kingdom as those able to make requests. Because Wikimedia used a manual contact mechanism, nothing prevented a resident of any other country from submitting a request, though the provided email addresses reflected the expected geographies (EUrepresentative.Wikimedia@twobirds.com and UKrepresentative.Wikimedia@twobirds.com). It is unclear what, e.g., California or Brazil residents legally entitled to request their data from Wikimedia were expected to do, since the privacy policy made no mention of them.

When collecting workflows, researchers answered questions about their residence honestly and did not make requests when instructions indicated they were ineligible. As such, we were not able to systematically collect denial data, but we did encounter eligibility checks and subsequent denials at different points in the request process. In Walmart’s dedicated DSAR form flow, the first question was about the consumer’s U.S. state of residence. Two researchers from ineligible states were directed to a screen ending the survey: “additional requests [besides tracking opt-outs] are not available in your state of residence.” One researcher was able to make a DSAR request to the Washington Post via their dedicated form, but after verifying the request through an email exchange, was informed that the platform “currently only take[s] data access requests if you are a resident of the EEA, UK, [...]”

While we observed some platforms built location-based logic inside a form, five did so by

The form for Alabama residents includes a 'State' dropdown set to 'Alabama', an 'I am the' section with 'Data Subject' and 'Parent/Guardian' options, and a 'Select request type(s)' section with 'Request to Delete Personal Data' and 'Video Tracking' options. The 'Request to Access Personal Data' option is not visible.

(a) An Alabama resident can make a data deletion or “video tracking” request, but not an access request.

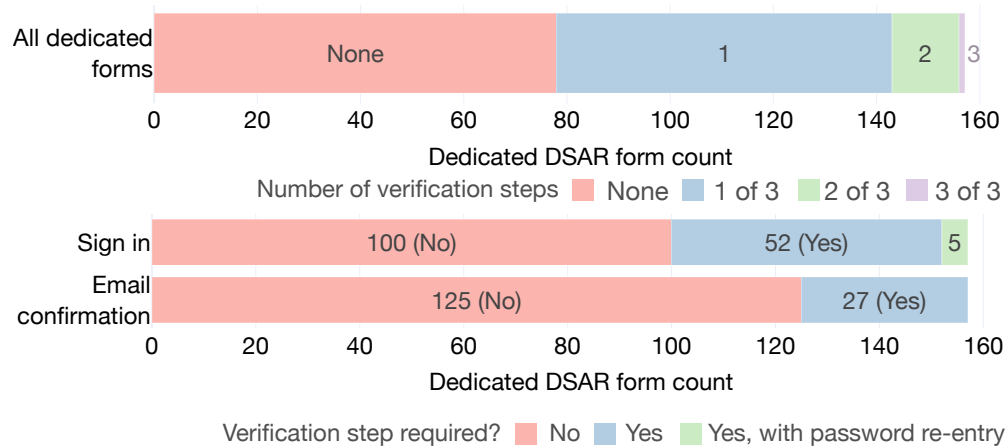
The form for California residents includes a 'State' dropdown set to 'California', an 'I am the' section with 'Data Subject', 'Parent/Guardian', 'Authorized Agent', and 'Business / Advertiser' options, and a 'Select request type(s)' section with 'Request to Delete Personal Data', 'Request to Access Personal Data' (shaded), 'Request to Correct Personal Data', 'Request to Limit Use & Disclosure of Sensitive Personal Information', 'Lodge an Appeal', and 'Video Tracking' options. The 'Request to Access Personal Data' option is highlighted with a yellow background and a dotted border.

(b) A California resident filling out the form can identify themselves more precisely and can make additional data requests (both represented by the dotted lines), including the access request (shaded), compared to an Alabama resident.

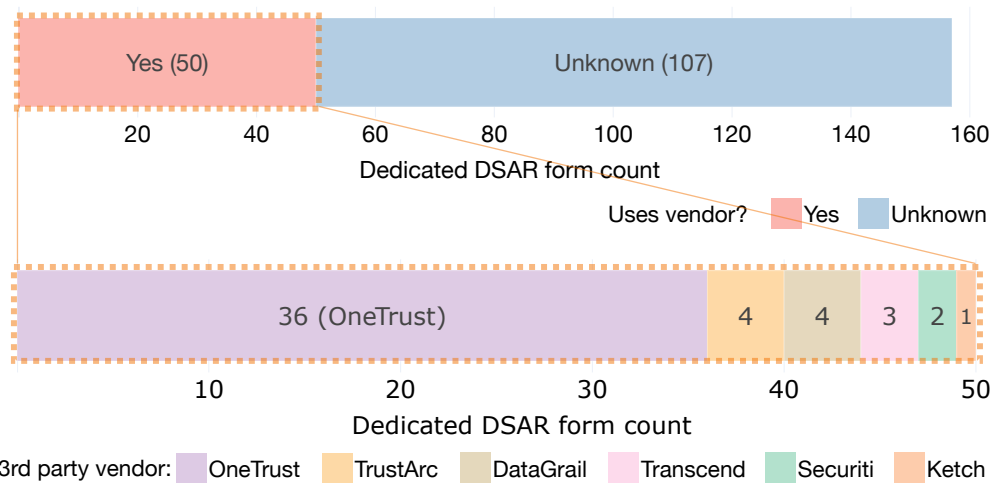
Figure 3.5: Details from USA Today’s dedicated DSAR form that has location-dependent behavior. <https://privacyportal.onetrust.com/webform/5b4b73e5-9c9c-4c5c-82c1-9389c9e42f1c/2f3a58de-6303-40ae-92ff-df56130329?anonymous-id=c272d355-c1df-4138-b2a3-c8c42e2acd7e&client-id=1d41cb36-428c-4e10-b882-05e24720d591>

creating parallel forms (f5; McDonald’s; Major League Baseball, MLB; Roadside America; Time Magazine). Among these, four had exactly two versions of their form: usually a specialized one (e.g., for the E.U.) and a general one. MLB, however, had 13 forms, each one corresponding to a U.S. state with privacy legislation.

Three of these platforms used a third-party compliance vendor: MLB and f5 used OneTrust and Time Magazine used Ketch. However, other platforms using OneTrust built location-based logic into a single form (e.g., Reuters’s form opened with a residency question that dictated the options for every further question). Sometimes one platform’s parallel forms were surprisingly dissimilar. f5, a U.S.-based cloud application and security company, phrased similar concepts in different ways in different forms. In its California-specific form, e.g., the consumer could self-identify as an “Employee of a Customer” or “Employee of a Partner,” whereas in the location-agnostic form the options were simply “Customer” and “Partner” (see Figure 3.4).



(a) The prevalence of the three different ways platforms verify consumer identity when they are filling out a dedicated DSAR form and overall.



(b) The top figure shows the platforms with dedicated forms that use a third-party vendor to aid in processing a DSAR. The bottom figure shows these vendors.

Figure 3.6: Authentication mechanisms and providers for the platforms in this study.

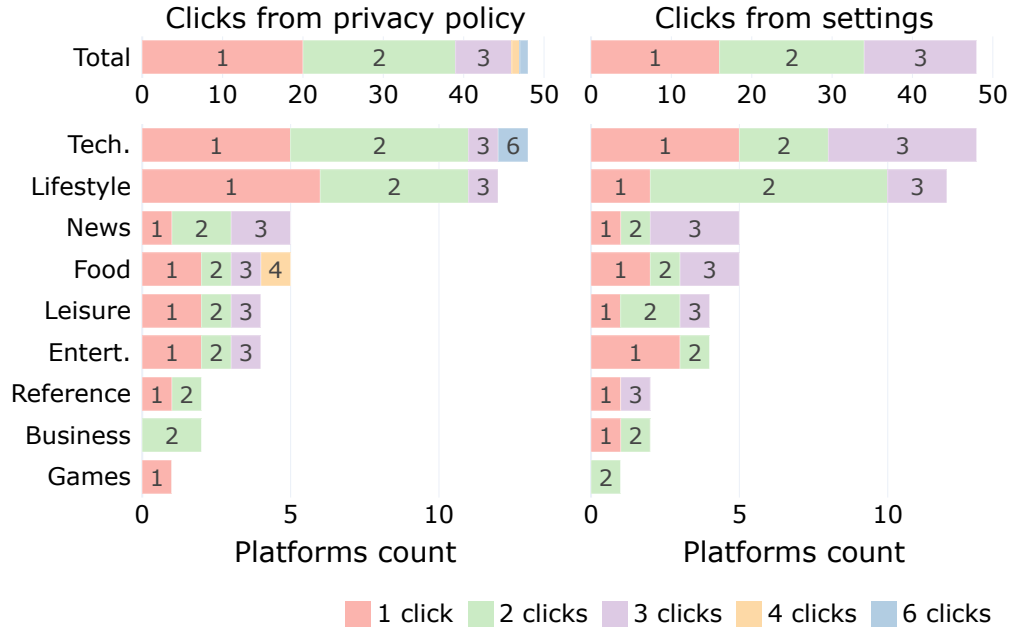


Figure 3.7: The minimum number of clicks to arrive at the DSAR form from the privacy policy and landing page of each platform. Per the legend, the number inside the bar-plots refers to the number of clicks rather than the number of platforms.

3.4.3 Discoverability and Obstacles

Before requesting their data, consumers must first locate where and how they are able to do so. In this section, I discuss factors that impact the availability of mechanisms and ease with which they can be found. To measure discoverability, we counted the minimum number of clicks required to reach the DSAR from the privacy policy page of each platform (dedicated or generic forms only). For platforms whose DSAR mechanism was embedded in user settings, we also counted required clicks originating from the settings landing page. Given that manual contact mechanisms often only involved instructions in the privacy policy, I excluded them from this analysis.

Figure 3.7 shows that, for nearly all platforms, making a DSAR required at most three clicks. Starting from the privacy policies, required clicks (mean 1.6) typically took consumers to sub-pages or expanded elements within the policy to reveal a final URL to the form. Microsoft, the outlier, required a consumer to navigate through three pages and expand a

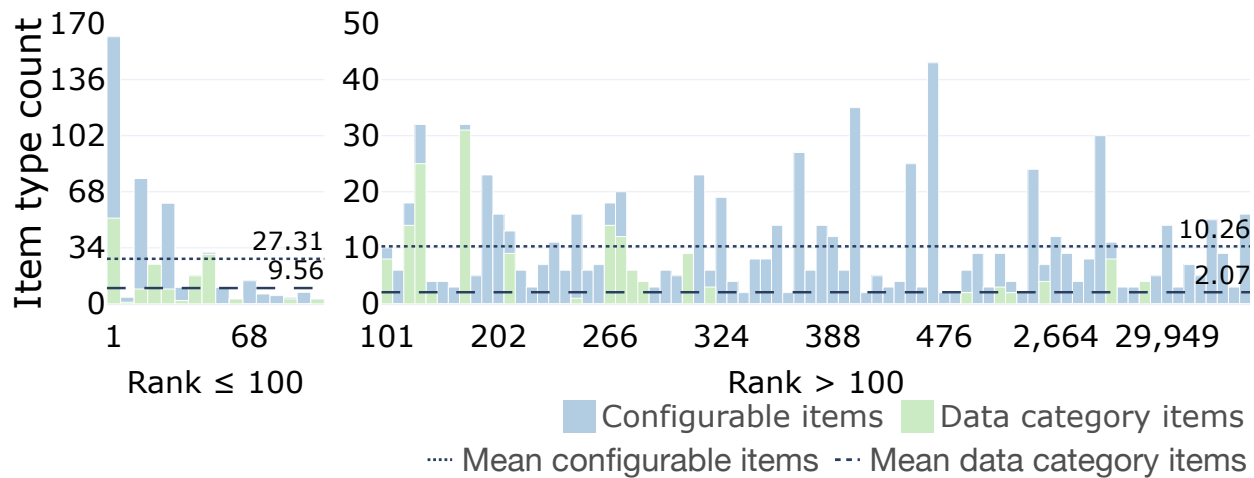


Figure 3.8: The number of configurable and data category items for 94 dedicated DSAR forms. Excludes 63 platforms with zero configurable items.

collapsible section, which contained the final URL. From the settings pages, a consumer experienced a similar trajectory, often requiring fewer than two clicks to reach the DSAR form (mean 1.8). Platforms usually included the mechanism in a settings section related to privacy or identity and used language such as “download,” “request a copy,” and “export” for the action and “personal information,” “data,” and “data export” to refer to the export.

Dedicated forms commonly included identity verification steps, which can impact usability. Some platforms required consumers to sign into their accounts, re-enter their passwords even if already logged in, or confirming ownership of their email address. Figure 3.6a shows that half of the platforms use at least one of these verification methods, with requiring sign in the most popular (36%). Some platforms, e.g., GitHub and Bitly, required little identity verification besides sign-in; others did so repeatedly and extensively. Platforms that used OneTrust to process DSARs had a characteristic, moderately heavyweight verification flow: a consumer fills out the form with their information (usually including a CAPTCHA), then must click a link sent via email to confirm the request, and then receives another email confirming the request is in process. Some time later (hours or days), the consumer receives another email from OneTrust on behalf of the platform, usually with the subject-line “Your

Privacy Request Needs Attention.” When they go to the URL linked in this email, they receive yet another email to the same address to confirm their identity, allowing entry into OneTrust’s portal. Now, after receiving at least four emails, the consumer can access the response to their single DSAR, which may be a request for more processing time or more information, or the option to download the data export itself.

The most extreme case of identity verification was that of Scribd, a U.S.-based digital document library technology company. In Scribd’s dedicated form [58], the consumer not only had to fill out typical information (e.g., name, email, and address), but also supply two separate “official proof[s] of residency, such as a driver’s license, government ID, passport, or other document that proves residence in the European Union” (even when the requester was not an E.U. resident).

Your Intuit products

☒ **TurboTax**
You'll get your tax return data, filing info, filing dates and return status, as well as your order history and personal profile data.

☐ **Mint**
You'll get your transaction and bill payment histories, budgets, financial goals, credit report, any customized offers we provided, and your personal profile data.

Data category items
Intuit

I am a (an)

Other Thomson Reuters Customer Thomson Reuters Former Customer

Thomson Reuters Employee Thomson Reuters Former Employee Thomson Reuters Job Applicant / Talent Community

Configurable items
Reuters

Date range

Last week ☐

Last month ☐

Last 3 months ☐

Last 6 months ☐

Last year ☒

Last 3 years ☐

All time
May take longer to export ☐

Date range options
Facebook

Edit file formats

Activity records HTML ▲

Timestamped records of your activity across different Google Products

Image attachments

HTML

JSON

File format options
Google

Choose a maximum file size that is most convenient for you to download. We'll divide your data into files of this size or smaller.

File Size

1GB

File size options
Apple

Figure 3.9: The five DSAR elements present in dedicated forms: configurable items, data category items, date options, file format options, and file size options.

3.4.4 Complexity and Configurations

So far, I have discussed who is allowed to make DSARs, the mechanisms for making them, and the steps a consumer must take to find these mechanisms. Here, I discuss the elements with which the consumer directly interacts and the decisions they make when making a DSAR. I only discuss dedicated DSAR forms since manual contact mechanisms and generic forms at most require a contact recipient, a subject line, and a message body. In Figure 3.9 I illustrate the components of a dedicated DSAR form we measured.

Configuring data categories. A **data category item** is a toggle, check-box, or radio button that lets a consumer include or exclude a type of data from their export (e.g., “Ads” data in Discord’s dedicated form, Figure 3.1a). Thirty platforms’ DSAR forms included such items. Categories were as broad as an entire product line, as with Intuit, which has Mint, Quickbooks, and TurboTax as data category items. Meta and Google had by far the most data category items, with as many as 58 and 60, followed by eBay at 31 and Yahoo at 30.

Google (rank 1 in Figure 3.8) was the only platform for which different researchers saw different numbers of data categories (60, 56, and 51). Only one researcher, for example, was shown the option to download “Business messages: Your messages with businesses,” likely because the other researchers had simply not exchanged messages with businesses through Google. This type of exclusion was not observed in other platforms (e.g., no researcher had used Intuit’s Quickbooks product, yet all were shown the option).

About half (16) of the 30 platforms with data category items defined these categories for the consumer, either in the form itself or via a link with more information. Others provided no explanation despite potentially confusing category names; eBay, for example, offered “Payment User Info” and “Payment Order Info” without explanation.

Of the 30 platforms with data category items, only six were not in the top 300 Tranco websites, and we identified only five that used a third party service. As with other cases, a

Table 3.4: Configuration features and how present they are in dedicated DSAR forms. Note that very few support full configuration of the features.

Feature	No. platform supporting		
	Fully	Partially	No support
Export within date ranges	6	2	149
Custom file formats	6	7	144
Repeated, automated requests	0	1	156
Max ZIP file size	0	5	152

platform with a high Tranco rank has a substantial consumer base and a product scope that may allow them to build a more fine-grained DSAR experience.

Other configuration features. I define **configuration items** as any options that can be selected or toggled in a form, including, but not limited to, data categories. One common example asked the consumer to identify their relationship with the platform (e.g., “Thomas Reuters Customer,” “Thomas Reuters Employee” in Figure 3.9). I exclude location items (e.g., a menu item for a consumer to pick the country in which they reside) from the definition of selectable items, since a consumer would not have to read every country or state listed to know which to choose.

Other configuration items we discovered were: specifying dates when the requested data was generated, the format of the files in the export, maximum sizes for the compressed files, and the ability to generate automated and recurring requests. The support for these items, in particular, fell on a spectrum. Some platforms provided full configuration, such as Snapchat, that let consumers request their personal data generated within any two dates. Other platforms only provided partial support, such as Spotify, which let consumers pick their data from either the past year or all time. In Table 3.4 I show that few platforms support full configuration of these features (the blue, shaded column). In fact, 88% of platforms supported none of these at all and only one platform, eBay, fully supported more than two (dates and format).

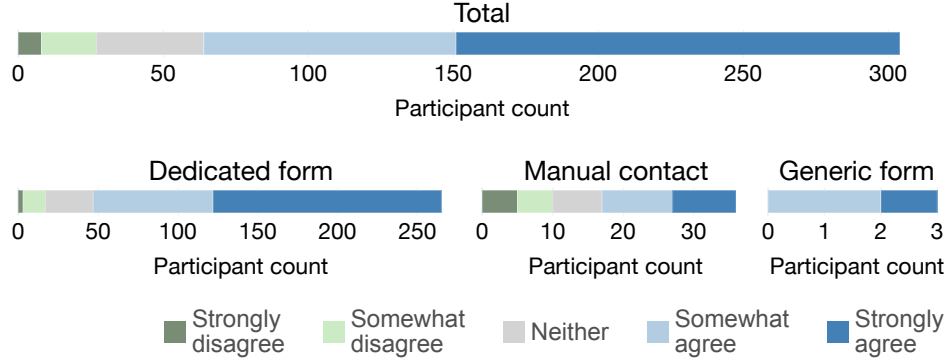


Figure 3.10: Degree to which participants agreed with the statement “The process of requesting my data was easy” for all requests and grouped by mechanism.

3.4.5 Participants’ Reactions

In the previous sections, I reported on the data collection and analysis completed by the research team on the request portion of the DSAR process. In this section I describe the results of the participant study where we collected data exports as it relates to requests. After asking participants to make DSARs, we asked them what they thought about the process. Figure 3.10 shows that most participants had little trouble: only 9% found the process difficult. Among favorable participant answers (215 total), 94 attributed this to the low effort or small number of steps involved in the request. The second most highly-cited reason (79) was the clear instructions we provided to participants making requests, which may indicate that additional guidance in making DSARs was key in their positive experiences. Table 3.2 shows these reasons along with their prevalence.

Ten participants were unable to submit requests. Among these, three had problems with submitting the DSAR itself (the others had procedural problems, e.g., misunderstanding instructions). One participant tried to submit a request to CapCut (video editing app owned by TikTok’s parent company, ByteDance), but could not pass the audio-based CAPTCHA on the form. After downloading the audio file, the participant had to listen for letters and enter what they heard. Unfortunately, after attempting two separate audio files, they gave up because the sound quality was “trash”: “I couldn’t tell if it was saying Z or C. I or E.”

The remaining two participants (submitting to Disney/Disney+, and AllTrails) experienced technical problems with the DSAR page. The vague error messages (“An unexpected error occurred,” AllTrails; and an invalid data message from Disney+) were not enough for them to troubleshoot the issues on their own.

3.5 Discussion

Based on our analyses of the DSAR request process and the data exports platforms return in response, I discuss key impacts on consumers.

3.5.1 Variability’s Impact on Consumers

One of our recurring findings was the high degree of variability across platforms in both the DSAR request process and the data exports returned in response. Regarding requests, this variability means that consumers who have learned how to make a DSAR request on one platform might have to adopt completely different strategies for searching for the appropriate request mechanism on another platform. In fact, the request mechanism itself might be completely different (e.g., structured online form vs. free-form email).

Differences in how platforms enforce geographic restrictions may also be confusing for consumers. Some platforms provided all consumers, regardless of jurisdiction, the same data access capabilities. Others forced consumers to search for different request channels, or to have no opportunity to make a DSAR request, based on their jurisdiction.

That the structure and contents of request mechanisms differed so much across platforms also impacts consumers. A consumer who has learned how to navigate one platform’s request interface might have to learn new terminology (e.g., the names of data categories and configuration items) and new directory authentication flows.

That said, the emerging use of third-party compliance vendors has the potential to better standardize the experience for consumers. For instance, if a few third-party compliance

vendors each implement the DSAR functionality for many different platforms, one could imagine this market consolidation could aid the development of tools that request a given consumer’s data from many different platforms. Furthermore, it could potentially aid data portability since single entities would be positioned to understand how to map data from the format of one client to the format of another client. Unfortunately, this potential remains mostly aspirational. Furthermore, some compliance vendors move the DSAR process outside the consumer’s existing workflow on a platform (e.g., forcing them to leave an authenticated account and provide other documents to verify their identity).

3.6 Conclusion

In this chapter, I characterized the full Data Subject Access Request (DSAR) process by analyzing and documenting interfaces from 250 popular online platforms. I define a taxonomy of this request process, developed from the analysis of the 250 platforms. We found that platforms have surprisingly variable approaches to implementing this right of access. Those with higher consumer traffic tended to have more configurable and complex request mechanisms, such as allowing consumers to choose the file formats for their data. We found that, in addition to varying in their complexity (e.g., the number of check-boxes in a form), DSAR mechanisms varied in their ease of access and availability (e.g., number of clicks from a privacy policy page to the DSAR) and configurability (e.g., request for data within a date range).

CHAPTER 4

DESCRIBING THE DSAR DATA EXPORT

4.1 Introduction

As I describe in Chapter 1, previous work has found that the request process and personal data consumers get back from making Data Subject Access Requests (DSARs) is not the most easily accessible [19, 173]. Moreover, it is not known how implementations of this right, specifically how the data exports are generated, vary across companies and across regions with differing legislation. In this chapter, I present a study that measures and describes how 69 platforms construct and organize data exports.

While details about a **Data Subject Access Request** (DSAR) process vary by jurisdiction, the GDPR was the first large-scale law to guarantee this right and, therefore, is a strong reference point for many other pieces of legislation that have come since [81]. The GDPR and most corresponding laws in other regions specify that the content of the data export must be in an “intelligible and easily accessible form, using clear and plain language, in particular for any information addressed specifically to a child” [49]. Additionally, the data must be “concise” and “structured” in a “commonly used and machine-readable format.” What constitutes “personal data,” though, is less agreed upon. While the GDPR defines it as “any information relating to an identified or identifiable natural person,” which includes a data subject’s “name, an identification number, location data, an online identifier,” this still leaves room for flexible interpretations.

Even with the guardrails provided by legislation like the GDPR, many terms are left undefined (e.g., “accessible form”) and procedures are vague (e.g., through what mechanism should data controllers send a data export?). This may result in data exports that are structured and created in different ways for different platforms. Past work has identified the barriers consumers face in comprehending the exports themselves, with a sample of social

media and e-Commerce platforms [132].

In analyzing these exports, we asked: On what axes can we compare data exports from different platforms, and how do they differ along these axes? We sought answers for this research question by conducting an online survey-based measurement study open to participants from the 19 US states and 105 countries with active legislation guaranteeing the right of access at the time of the study [3, 122, 81, 77, 55]. Participants donated 117 de-identified data exports from 69 platforms. We analyzed raw data exports from the research team together with the de-identified participants’ exports to produce our primary contribution: **a public repository with anonymized, generalized data export schemas and export format specifications (e.g., file formats and directory structure) for these 69 platforms.**

4.2 Background and Related Work

Data exports, collections of files containing personal data obtained from making a Data Subject Access Request, are used for many reasons other than just as a tool for consumers to enact their privacy preferences. Work in a variety of disciplines has leveraged data exports for purposes beyond data subjects’ rights [65, 175, 183, 172]. For instance, researchers have used data exports (as “data donations”) to audit social media platforms like Instagram [137], TikTok [185], and Twitter [177]. Given the wide applicability of data exports, it is important they are well-characterized and understood, which past work has covered as well. Two studies, for example, highlighted vulnerabilities in the DSAR authentication process that can be abused to steal data from others [16, 106].

Past work which explored data exports indicated that they may often be incomplete and difficult to use for consumers. Researchers analyzed the data exports generated by five large, online platforms and showed how the amount of advertising data varied significantly among those platforms [94]. In a follow-up study, they compared data exports made in

2023 to those made in 2018 (when the first study took place), and found that companies were likely under-reporting data and data categories during their first study [131]. Li, et al. similarly highlighted that among Android-based mobile apps based in China and the U.S., many may be withholding data from consumers given the mismatch in records in exports compared to those listed in privacy policies and based on the APIs used in those apps [95]. Finally, other work has shown that while encoding data in machine-readable formats may be useful for transferring data between platforms, is not readily usable by the average consumer, especially given a lack of data definitions and descriptions [19, 161].

This work builds upon these studies in two key ways. First, rather than analyzing request experiences and data exports from accounts that were synthetic or owned by researchers, we recruited participants who were active users of the platforms in our study. This helped us achieve breadth and reliability in our data, rather than the more limited sample set made up of synthetic user data or constrained to requests from the country in which we reside. Second, the Data Export Schema Repository I am releasing as part of this work is scaled to a much larger number of platforms (69) than the one existing data set we found (previously five). Additionally, I am making it open-access to promote the transparency and utility of the file types, directory structures, and data schemas within these exports for academics, researchers, tool-builders [76], and internet users who could benefit from it.

4.3 Methods

4.3.1 Data Export Collection

We relied on data donations for our repository of data exports.

Participants. We recruited participants from the crowdwork platform Prolific and, to supplement data for the export portion of the study only (Section 4.4), from our research group. Participants from both sources needed to speak English, be at least 18 years old,

use at least one of the platforms we pre-selected for the study for at least one year, and (for Prolific participants) have over 10 completed tasks with an acceptance rate over 95%. Participants also had to live in a place with active legislation granting them the right to access their personal data. We used the International Association of Privacy Professionals’ U.S. state level [55] and country level [77] tools to identify regions that granted the right of access free of charge [3, 81, 122].

Procedure for Prolific participants. Prolific participants who completed the study took at least three surveys. In the first, they identified their country (and, if applicable, state) of residence and reported which of the 69 platforms they actively used. Participants who had used platforms to which we had received few responses and passed the attention and automation checks in the survey were then passed onto the next survey, where they were given instructions to make DSARs to any of the eligible platforms for which they were willing. Three days after making the requests, participants were given access to the submission survey, where they uploaded their data exports to a web-app we developed for this study. Participants were compensated 3 USD for the first survey, 1 USD for each platform to which they made a DSAR, and 2 USD for each platform whose export they submitted.

Procedure for additional research group participants. To collect additional samples of data exports, we condensed the procedure above and reached out to collaborators of our research group to invite them to participate. In the condensed version, we only asked participants about the platforms they used, gave them instructions to make DSARs to any of those platforms, then sent them the link to our web-app for data export submission. Due to the condensed nature of this study, we did not compensate this group of participants.

4.3.2 *Protecting Participants’ Privacy*

We developed a data processing procedure in TypeScript to de-identify every data export the participants donated to the study. This procedure ran locally on each participant’s browser; meaning the original files containing their personally-identifiable information (PII) never left their machines. The procedure first filtered out the content of non-text files, such as images and videos, only retaining their names, relative paths, and sizes. It then parsed and de-identified text-based files differently based on their file extensions. When de-identifying a structured file (one with JSON, CSV, XLSX, or HTML extensions), the procedure retained all column names, keys, and HTML tag metadata and de-identified the rest of the text using the two techniques described below. For unstructured files (i.e., every other non-media file type, such as TXT or MD), the whole file was de-identified using both methods.

Technique 1: Character replacement. In the **character replacement** procedure, each alphabetic, ideographic, and syllabic character (including Emojis, Kanji, and Chinese characters) are each replaced with a `*` and each digit is replaced with a `#`. Files de-identified using this technique allowed us to analyze the approximate layout of files and identify instances of record types, such as phone numbers (which could appear as `###-###-####`) and emails (`****@*****.***`).

Technique 2: Hashing with salt. In the **hashed** procedure, the script de-identified each term separated by symbols or whitespace separately. Each term was prefixed with a 128-bit **salt** using the industry standard UUID protocol, implemented in the `uuid` NPM package [41, 83]. Each salt is specific to a participant-platform pairing (i.e., participant A on Google has a different salt than participant A on Meta). The salt ensures that a single term appearing in exports from different participants has digests unique to each participant, preventing the research team from identifying links between the participants’ PII. The script then hashes each salt-term string with the SHA-512 function.

The research team had no interest in identifying collisions or gleaning PII from participants’ exports. Even if we did, it would have taken about 200 days to find a single collision for any given digest using the most powerful resources available today. An NVIDIA GeForce RTX 5090 computes about 10^{10} hashes per second. De-identified files were sent to our research group’s access-controlled server over HTTPS, where they remained. After completing our data analysis, we deleted all contact information for non-Prolific participants and every file we received.

4.3.3 Ethical Considerations

In this section, I report on the stakeholder-based ethics analysis we conducted to evaluate the studies in this chapter. I do so by listing the different stakeholders we identified and, for each one, articulating the harms we considered, the mitigations we developed against these harms, and how our procedure and decision to conduct the studies ultimately upheld the four principles in The Menlo Report [169].

The participants I described in Section 4.3.1 had to complete at least three surveys, interact with and make DSARs to platforms they are regular users of, and ultimately submit a de-identified version of their data export, which likely contained personally-identifiable information (PII) before de-identification, to us. As an additional mitigation to all the harms I describe below, we received approval by our university’s Institutional Review Board, to whom we submitted our study procedures, before conducting the study. Additionally, researchers who interacted with participant data completed our university’s Human Subjects Protection course. Finally, we made it clear to participants in our study materials that they were able to opt out from the study at any point and still be compensated for their participation up to that point (Respect for Persons and Justice).

Harm: participant PII is obtained by an adversary with access to the participants’ browser or network traffic. As we were the developers of the web-app participants used to submit their data exports, we took care to ensure the tools and code we used did not rely on network communications with a third party. We also ran our compiled web-app through NPM’s vulnerability auditor and found no immediate threats [80]. Additionally, we configured our web-app to only support HTTPS requests. Finally, any data from the data export was deleted from the participant’s browser’s IndexedDB when the de-identification process was complete. We believe these mitigations, which are standard practice on the web, are enough to significantly mitigate this harm (Beneficence) and were applied to all participants (Justice).

Harm: participant PII is obtained by the research team. The primary mitigation against this harm was the de-identification mechanism I describe in Section 4.3.2. Given the risk of PII being found in file, directory, and record family names, we took several precautions to minimize the number of researchers with access to this PII. First, the server where we stored the de-identified data exports was only accessible to a subset of the research team. One researcher then ran a script (described in Section 4.4.2) to extract schema-generalized file paths and record-family names in the export. If any PII were found, they were replaced with a schema-generalized version of that string that stripped PII and the schema generalization script was adapted to prevent any future PII of that type from being not being removed. We considered these sufficient precautions to prevent against this form of leakage.

Harm: participant PII was inadvertently included in the Data Export Schema Repository. In addition to following the previously mentioned mitigations, we also manually looked through every directory, file, and record family name multiple times before publishing the repository, to ensure that only general, non-identifiable data was included in the repository.

4.3.4 Limitations

We experienced significant participant dropout during the study. From a pool of 344 participants, 195 passed the quality checks in the screening survey. All were invited to move onto the next stage of the study, but only 195 made at least one request and 27 of those submitted at least one export. We reached out to participants who did not make requests or submissions despite the opportunity, but most among those who responded had forgotten about or given up on the study or claimed they never received their export. Future work could explore alternative methods for recruiting participants, possibly focusing on those who would be excited to participate in a study about online data privacy.

To gain an even deeper understanding of the content of data exports, we experimented with using locally run machine learning models and LLMs to classify records based on keys (in JSON files) and column names (spreadsheet files). Despite our attempts, no model performed with more than 50% accuracy. We presume this is because many JSON keys and spreadsheet column names in our export are too vague or generic to be accurately labeled, even by a human.

Finally, it is important to highlight that the analysis on requests and exports are representative of the moment in time in which that data was collected. I note, for example, that while collecting data on the Major League Baseball platform in mid-2025, consumers had 13 forms to choose from.

4.4 Characterizing the Data Export

In this section, I describe the **data exports** that study participants donated to supplement our exploration of Data Subject Access Requests. In Section 4.4.1, I describe several characteristics of data exports, such as their sizes, number of files, number of records, and general structure. Then, in Section 4.4.2, I describe the process of schema generalization we applied to files in the data export collected and introduce the Data Export Schema Repository.

Table 4.1: The composition of data exports the study’s participants donated. Note that the largest export size corresponds to the largest file size (shaded cells).

Measure	Mean	Median	Min	Max
<i>Export composition</i>				
Sizes (KB)	24,843	142	0.26	1,240,259
Files	135	6	1	6,300
File families	16	4	1	117
Directories	8	1	1	331
Depths	1	1	1	5
<i>File composition</i>				
Sizes (KB)	183	1.08	0	1,240,259
Records	3,800	88	1	116,440
Record families	189	67	1	1,884

4.4.1 Making Sense of the Data Export

To analyze data exports from a variety of platforms, we collected 117 unique data exports from 27 participants across 69 platforms (a subset of those from the study described in Chapter 3). As described in Section 4.3, we did not receive most of the content of participants’ exports, and therefore confine our analysis to their structure.

Table 4.1 shows that exports, like requests, were quite varied. Exports from platforms like Ticketmaster and Bit.ly had only one file each, while at the other extreme platforms like Strava and ChatGPT provided hundreds of files each and X provided thousands. A “typical” data export looks more like the median values in Table 4.1: around 140 KB, six files, and one directory. As Figure 4.1a shows, the most common file sizes were between 1 and 10 KB.

Platform-built user interfaces. Making sense of data exports, their structure, and what they contain requires different strategies for different exports. Among the many reasons a consumer may decide to request an export of their data, few, besides pure exploration, can be successful without a knowledge of the structure of the export.

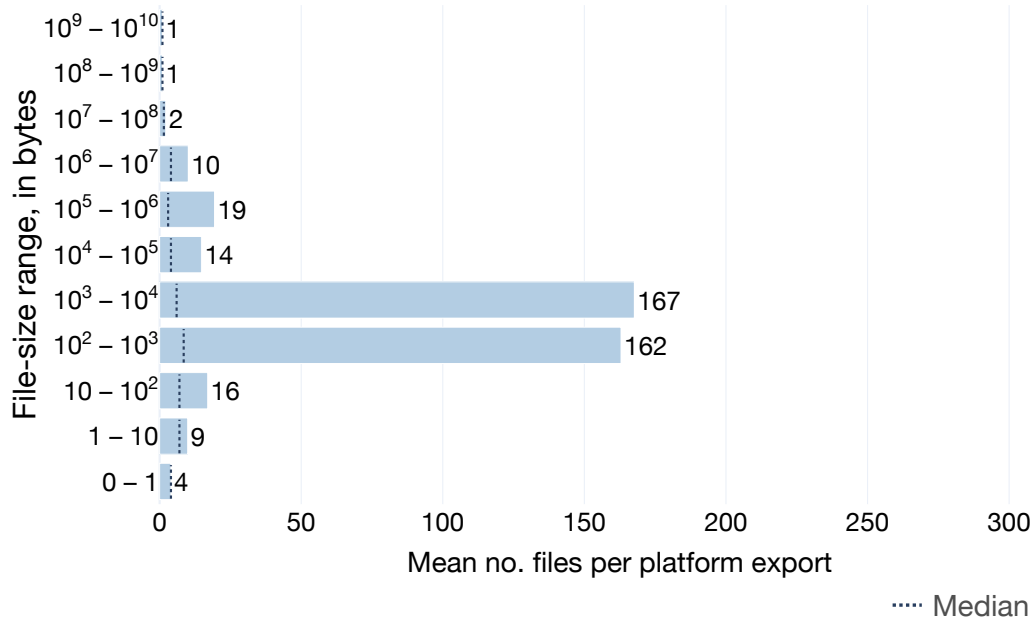
Some platforms have taken steps to make their exports more user-friendly. One such example is X (formerly Twitter), whose export structure was initially puzzling. The most common file format in the two exports we received for the social media platform was SVG,

an uncommon media file format for general purpose images. Even more unusually, both exports contained exactly 3,079 SVG files with identical names and directories.

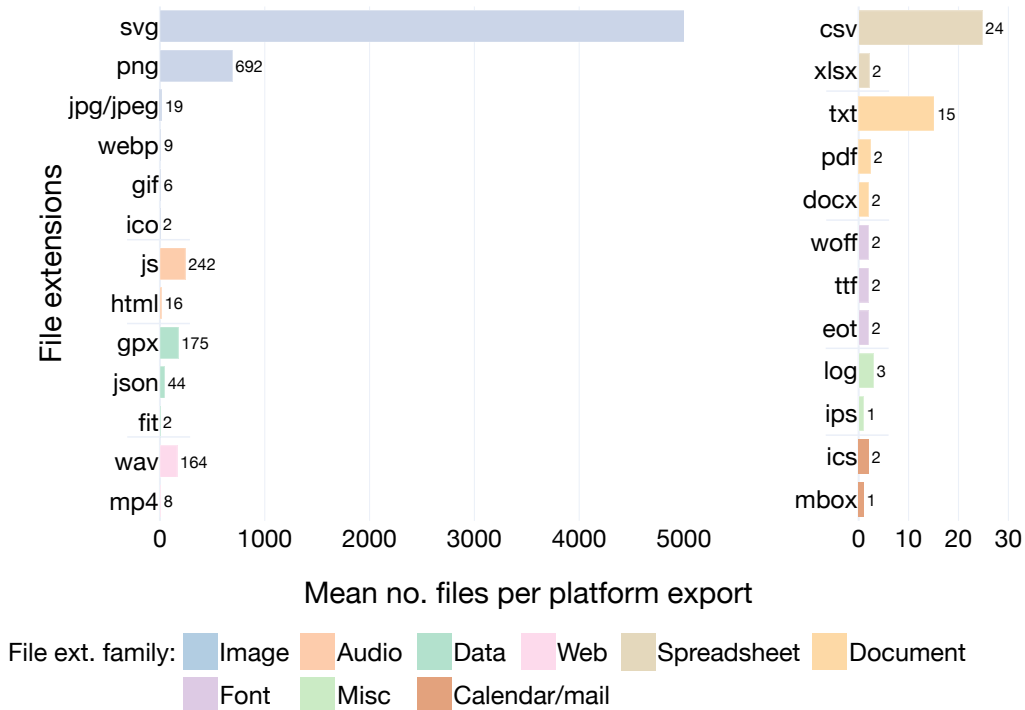
We realized that some platforms include files (in the case of X: HTML, CSS, and SVG) to help consumers interact with and comprehend their exports. These HTML files often contained JavaScript that read from other files in the export to provide consumers with an interface for interacting with their data. In addition to X, exports from 16 platforms had HTML files and, as Figure 4.1b shows, on average platforms did tend to have around three HTML files per export. Not all HTML files have the sole purpose of being data visualizers. Google exports, for example, had files in deeper directories with more specialized names (e.g., `Takeout/Chrome/Bookmarks.html`), which indicates these files may not only help consumers visualize their data, but store the data themselves.

Beyond HTML visualizers, we found that platforms like Amazon, Tumblr, and Yahoo included files dedicated entirely to showing the schema of their data export. However, of the 69 platforms from which we collected exports, only 18 (26%) included a schema, “read me,” or HTML file. The majority of platforms, therefore, likely lack significant guidance as to how the export should be used.

Contrasting approaches to data storage. In comparing data exports, we identified a tension in the strategy platforms chose to store consumer data, especially when it came to large amounts of data. To illustrate this, consider how different platforms stored conversation data. In one export, ChatGPT opted to store multiple conversations in a single `conversations.json` file containing about 17,000 KB of data. TikTok followed a similar strategy, seemingly storing all the participant’s conversations in one file (`DirectMessages/DirectMessages.txt`). In contrast, Rockstar Games’s Grand Theft Auto (GTA), a mobile-app game, provided 289 `messages.json` files averaging only 24 KB each, under different directory names. Instagram took the same approach, with different directories within `your_instagram_activity/messages/inbox/` appearing to belong to different conversations.



(a) For each export we received, the average number of files for each size bin, in bytes. Each bin is an order of magnitude.



(b) For each export we received, the average number of files for each file extension.

Figure 4.1: Characteristics of the 117 data exports we collected.

4.4.2 *Toward Standardized Exports*

As I discuss in Section 4.3, when participants donated their exports, most of the content of the files were not sent to us, except for spreadsheet files (i.e., CSV, TSV, and XLSX) and JSON files, for which we retained tab names, column names, and JSON keys. I discuss the content of these files in terms of **record instances** and **record families**. Consider a CSV file listing a consumer’s contacts with the columns: `first_name`, `last_name`, `phone`. If there were 10 rows in this file after the header, there would be 30 record instances (10 `first_name`, 10 `last_name`, and 10 `phone`). A record family, on the other hand, describes the type of data in that record instance. In this same example, there are only three record types, corresponding to each of the columns. The same distinction can be applied for the definitions of **file instance** and **file family**.

Data within export files varied greatly across platforms and, at times, between participants. Tumblr’s data export, e.g., included the record family `[] .data.crushes. [] . <blogname>`, where a consumer’s blog names were encoded in JSON keys (we use JSON dot notation to discuss JSON record families). This happened for file and folder names as well, as with Instagram, whose chats were grouped in separate directories of the format `your_instagram_activity/messages/inbox/<username>/`. To accurately describe the record families of different exports, we applied a process of schema generalization to file, folder, and record instances. That is, we replaced participant-, export-, or instance-specific text tokens with semantic placeholders. I discuss all the transformations we applied in the Open Science appendix. Figure 4.2 shows a sample schema we automatically generated using our approach for `account_settings.json`, one of many files included in a data export from WhatsApp, a communication app. On line 32, `blocked_#p_users` the `#` character was originally a digit and on line 50, `<number>` used to be a short integer.

The “family” figures in Table 4.1 were computed post-schema-generalization.

Case study: comparing LLM conversational agent data. To illustrate the barriers that make it difficult to translate data between exports as currently implemented, consider a consumer trying to combine their chat histories from two LLM conversational agents: ChatGPT and Google’s Gemini. To make this task as simple as possible, assume the consumer’s primary goal is to capture all messages exchanged with each conversational agent, including the author of each (the consumer or the agent) and associated timestamp. We further assume (as discussed in Section 3.4.4) that the consumer configured their Google DSAR to request only Gemini data. (ChatGPT has a more limited DSAR form, but I argue this is probably adequate given it provides much fewer products and features.)

Google’s export has three files with names mentioning Gemini; ChatGPT’s export contains five non-media files that may contain chat history. Even after identifying the correct files (possibly `Takeout/MyActivity/GeminiApps/MyActivity.json` for Google and `conversations.json` for ChatGPT), the consumer must identify the three relevant data points (message, author, time) among 475 record families from ChatGPT and six from Google. None of the record families in Google’s file seem to indicate authorship (`header`, `title`, `time`, `products`, `activityControls`, and `safeHtmlItem.[] .html`), whereas a few in ChatGPT’s file do (`mapping.<unique_id>.message.metadata.aggregate_result.messages.[] .sender` and `mapping.<unique_id>.message.author.name`). Ultimately, the consumer would need to thoroughly investigate the structure and values of each record to make progress toward their goal.

```

1  {
2    "$schema": "http://json-schema.org/schema#",
3    "type": "object",
4    "properties": {
5      "wa_settings": {
6        "type": "object",
7        "properties": {
8          "last_seen_privacy": {
9            "type": "string"
10         },
11         "profile_photo_privacy": {
12           "type": "string"
13         },
14         "about_privacy": {
15           "type": "string"
16         },
17         "status_privacy": {
18           "type": "array",
19           "items": {
20             "type": "string"
21           }
22         },
23         "blocked_numbers": {
24           "type": "array",
25           "items": {
26             "type": "string"
27           }
28         },
29         "blocked_numbers_update_time": {
30           "type": "string"
31         },
32         "blocked_p_users": {
33           "type": "array"
34         },
35         "read_receipts": {
36           "type": "string"
37         },
38         "is_dma_eligible": {
39           "type": [
40             "boolean",
41             "integer"
42           ]
43         },
44         "opted_in_dma_integrators": {
45           "type": "array"
46         },
47         "settings_last_fetch_time": {
48           "type": "object",
49           "properties": {
50             "<number>": {
51               "type": [
52                 "integer",

```

Figure 4.2: An example schema automatically generated for WhatsApp's `account_settings.json` export file. Lines 32 and 50 show instances of schema generalization.

4.4.3 *The Data Export Schema Repository*

Given the high variability between exports, the breadth of files and data within them, and the complexities that arise at every step of the Data Subject Access Request process, we introduce the Data Export Schema Repository (<https://anonymous.4open.science/r/dsar-export-schemas/>). This repository holds files and folders that contain critical information about the exports of 69 of the platforms in this study. Each top-level folder identifies a different platform. Within those are the folders and file families that a consumer would see if they were to make a DSAR. We generated schemas, using the JSON Schema standard [180], for most spreadsheet and JSON files, except those for which we were not able to consistently and reliably apply schema-generalization. All of these schema files are suffixed with `.schema.json`. Other files in the repository have no content, but are included to show all file families, not just those in machine-readable formats, to maximize the repository’s utility for all uses. E.g., researchers who have export schemas in advance of collecting their data can plan analyses and procedures without having to wait for data donations.

I have published our Data Export Schema Repository anonymously at the following URL: <https://github.com/UChicagoSUPERgroup/dsar-export-schemas>. The repository contains information about the structure and layout of data exports from 69 platforms. Each export is in a directory under `schemas/` named after that platform (e.g., `Airbnb` and `Too_Good_To_Go`).

I will use AllTrails’s directory to illustrate how to navigate an export schema in this repository. By navigating to `AllTrails/` (<https://anonymous.4open.science/r/dsar-export-schemas/schemas/AllTrails/>), you are viewing a simplified and generalized version of what a participant would see in their data export. With the exceptions described below, the names of the files and directories are as they appear in a real data export. Rather than trying to emulate the “average” export in the repository, we instead chose to prioritize breadth: even if a file was only present for one participant in our study, as long as we were

```

{
  "Universities": [
    {
      "name": 'Boston University',
      "enrollment": 36624
    }, {
      "name": "Brandeis University",
      "enrollment": 5302
    }
  ]
}

```

Figure 4.3: In this example (purely illustrative), there are four individual records and two record families. `Universities.[] .name = "Boston University"` is one of the four records and the two record families are `Universities.[] .name` and `Universities.[] .enrollment`.

able to schema-generalize the file, we decided to include it in the repository.

Files Without Content. In `AllTrails/` there are seven directories and one file (`remote.html`). That file has no content, but we still included it in this repository since it existed in one of the exports we received for AllTrails. It has no content because we were not able to schema-generalize HTML files due to their more complex structure, compared to spreadsheet and JSON files (which we did schema-generalize). We made the decision to include it, though, since it contains valuable contextual information: that AllTrails has an HTML file that may provide consumers with an interface for interacting with their data.

Interpreting Schema-Generalized Files and Records. We took steps to remove participant-specific information from file names and record names. This was not only to protect participant privacy, but to avoid communicating redundant information. Consider the lone `<custom_name>.jpg` file in `AllTrails/UserOnlineActivity/photos/`. Here, `<custom_name>` not only communicates that the file name does not follow a clear or useful naming structure that we could identify, but it also means that there can be multiple of this specific file type in this directory. In fact, **any time a schema generalization artifact is**

in a file or record name, it may occur more than once in that location.

Below is a comprehensive list of schema generalization artifacts. Unless otherwise noted, it can apply to directories, files, and records:

- `\#` : a digit.
- `YYYY` or `YY` : a four or two digit year.
- `MM` or `MMM` : a month expressed with digits or with by abbreviated name.
- `DD` : a day in a month.
- `HH` , `MM` , `SS` : hour, minute, and second, respectively.
- `<username>` : a username or handle in a platform.
- `<email>` : an email address.
- `<phonenumber>` : a phone number.
- `<unique_id>` : what is seemingly a randomly generated unique ID, such as a UUID [41].
- `<number>` : a specific case with JSON keys and spreadsheet column names (i.e., records) that are numbers.
- `<custom_name>` : the most general of the schema generalization artifacts, this is used when there is a consumer or export specific string that did not have a discernible pattern.

Schema Files. Some files in the repository have this additional suffix attached to their name: `.schema.json`. This can be the case for files of any extension, but for the 69 in this repository, they appear after JSON, CSV, and XLSX files. When this is the case, the content of the file contains the structure of the file represented as a schema following the

JSON Schema draft standard [180]. The content of JSON files are described by the schema in the typical way, i.e., “properties” list JSON keys and their value types. For CSV files, the schema “properties” correspond to CSV column names. For XLSX files, top-level “properties” correspond to sheet names within the file and second-level “properties” correspond to column names within that sheet.

4.5 Discussion

Based on our analyses of the DSAR request process and the data exports platforms return in response, I first discuss key impacts on consumers. I then describe potential uses for the data export schemas we developed, as well as future work.

4.5.1 Potential Uses For Data Export Schemas

As I presented in Section 4.4.2, the Data Export Schema Repository we developed is a public resource containing the directory structure, file formats, and file schemas for the 69 platforms for which we received donated data exports in this work. Below, I provide a set of recommendations for possible uses of the repository for two key stakeholders.

Researchers. One of the key ways we envision the Repository being useful to other researchers is by providing insight into the types of data they can expect to collect from different platforms in research studies involving donated consumer data. This utility can inform two distinct stages of the research process. The first is the planning stage of a study. A researcher might be curious whether a particular platform’s data export seems to contain a particular class of data, yet not have an existing account on that platform. Alternatively, they might want to search for a particular record type across many platforms. Our repository facilitates both uses.

The second is the data-processing stage. Researchers are typically morally obligated to

protect participants’ privacy, yet want to analyze as much study-relevant data as possible within these bounds. Some studies that have relied on data donation have unfortunately encountered repeated issues when trying to de-identify participant data due to the uncertain structure of the data [171]. Other studies have opted to do data processing on a remote machine (other than that of the participants), which is far from ideal for participants’ privacy [137]. In both cases, researchers knew which data they wanted to collect and which data they did not, but stumbled when needing to parse records they did not expect to find in the export [171].

Equipped with our repository, researchers who want to conduct similar studies in the future will be able to know exactly which files and records in those files to collect during the donation process, as well as how to parse the data for de-identification purposes. This can help with providing clear instructions to participants for making DSARs, especially when the relevant platform has a dedicated form with data category options to reduce the size and scope of the data donated. An automated system (e.g., a script sent to participants or a web app like the one we built) would be responsible for processing less data, meaning it can run more efficiently on participants’ own machines, increasing participant privacy.

Engineers and product designers. As I discussed several times in the findings of this work, data exports are highly varied between platforms. This is not surprising since different platforms have different data models and use different data stores for their customer data. Note, however, that regardless of the original data model, a translation step must occur between that schema, the storage mechanism (e.g., a relational database), and the files included in the export. Currently, this translation step follows loose guidelines set by laws like the GDPR (e.g., it must be “accessible” and “intelligible” [49]).

Rather than arbitrarily translating this data, I encourage engineers to explore paths taken by other platforms and learn from those formats. Some exports can serve as positive examples, such as by using HTML files to facilitate non-technical consumers interacting with

their data. They can also serve as constructive examples, by showing the consequences of ambiguous file names (e.g., `messages.csv` and `messages_archive.csv`) and storing several uncategorized files in the same directory. There is a clear opportunity for platforms and compliance vendors to set the standard for usable and transparent data exports.

4.5.2 Future Work

Future studies could expand on the data export analysis we started in this work. A primary limitation in our experimental set-up was the difficulty in obtaining information about the types of records in the data exports. Record families alone could not tell us about the values of those records for us to make claims about the data categories in the files. Building on the Data Export Schema Repository we developed, future work could deeply analyze files and record families of interest by processing records locally (i.e., on participants' machines). A core contribution of the schemas I release is to enable researchers and privacy practitioners to write code that successfully parses data exports from a specific platform without ever needing to see an example data export from that platform. Furthermore, by examining the semantically meaningful parts of the schema, researchers and privacy practitioners can identify aspects of a platform's data export that deserves further attention from the community.

One possible analysis is for a privacy researcher to run a consumer's records through data classification models, comparing those labels with the data categories platforms note they collect in their privacy policies.

An additional direction for future work that is also aided by the Data Export Schema Repository is tracking changes in the composition or structure of data exports over time. Such an analysis could provide temporal insights into the data collection and data use practices of different platforms. A similar experiment could involve evaluating different schemas used by a platform to store the same type of data by running a user study where participants are shown both options.

4.6 Conclusion

In this chapter, I characterized the data exports originating from Data Subject Access Requests (DSAR). 27 different participants donated 117 data exports from 69 distinct platforms. Similar to the request process (Chapter 3), we found that the exports varied significantly in directory structure, file types, and records across platforms. Platforms with higher consumer traffic tended to have larger and more complex data exports.

Beyond simply characterizing the data exports, we developed a method for abstracting each company's data exports into an abstract schema. To supplement our findings, I am publicly releasing the Data Export Schema Repository, which contains the directory structure and file schemas for the exports of the 69 platforms for which we received donated data to date. I expect these artifacts will better enable other researchers to analyze platforms' data exports in a more scalable way.

4.7 Additional Materials

4.7.1 Communicating Participant Privacy

The following text was included in the website participants accessed to donate their data.

Privacy Reminders

We will **not** receive your files in their "raw" form. Instead, for each file, we receive two modified versions.

Hashed file

We use the cryptographic techniques of salting and hashing to remove any personally identifiable information from your files. Hashing converts any text to a jumble of characters that retains no meaning from the original text. Salting makes it so that if that text appears multiple times within your submission, we'll

be able to see that it appears multiple times, but we won't actually know what's appearing multiple times.

Here's an example: **Raw Data (we do NOT receive this):**

```
1 {
2   "name": "Alex S.",
3   "year_of_birth": 2000,
4   "email": "s_alex@gmail.com",
5   "best_friends": [
6     {
7       "name": "Bob C.",
8       "year_of_birth": 2001
9     },
10    {
11      "name": "Bob R.",
12      "year_of_birth": 1993
13    },
14    {
15      "name": "Alice C.",
16      "year_of_birth": 2000
17    }
18  ]
19 }
```

Hashed data (we receive this):

```
1 {
2   "name": "ca978112 dcafafac23",
3   "year_of_birth": "ff8147c4", // also on line 16
4   "email": "8c883eaa",
5   "best_friends": [
6     {
7       "name": "b9807785a bbdcafafac",
8       "year_of_birth": "938c82e9"
9     },
10    {
11      "name": "b9807785a 231b39a23",
12      "year_of_birth": "10013cc3"
13    },
14    {
15      "name": "bb1b24322 bbdcafafac",
```

```

16         "year_of_birth": "ff8147c4" // same as line 3
17     }
18 ]
19 }

```

Character replacement file

So we can get a sense of the length and shape of the files you submit, we also create a version of each file with character replacement. In this method, we replace any alphabet character with the pound symbol (#) and any digit with an asterisk (*). We don't do any replacement of special characters.

We would transform the same raw data file example from above into:

Character-replaced data (we receive this)

```

1  {
2      "name": "**** *.",
3      "year_of_birth": "####",
4      "email": "*_*****@*****.*",
5      "best_friends": [
6          {
7              "name": "*** *.",
8              "year_of_birth": "####"
9          },
10         {
11             "name": "*** *.",
12             "year_of_birth": "####"
13         },
14         {
15             "name": "***** *.",
16             "year_of_birth": "####"
17         }
18     ]
19 }

```

Data we don't collect

We don't collect the file content from any non-structured text file. So we won't see any images, videos, etc.

Additional data collected

In addition to the pseudonymized versions of structured file types, we also collect:

- The directory structure of your data export, meaning which files are in which folders.
- The names of every file, including of non-structured files.
- The size of every file, including of non-structured files.

CHAPTER 5

DATA PORTABILITY AND CONSUMER-FACING DSAR DATA EXPLORATION TOOLS

5.1 Introduction

Several studies have leveraged participant data donations in the medical field [70, 135], information and communication sciences [22, 128, 14, 119], political science [57], and computer science [184, 29, 13]. This data comes from a variety of sources, but is mostly associated with the digital traces collected by platforms on those users. Since the passing of the GDPR [49], more studies have accessed this data via personal data exports obtained from making Data Subject Access Requests (DSARs), especially as online platforms have limited the features of and access to their public APIs [25]. Previous work studying data exports and data donations in research shows that there are, however, many privacy and security risks [28], challenges with study execution [30, 182], and data consistency and quality issues in using data exports for research [79].

While researchers have developed some tools for facilitating data donations, many are outdated, overly complex, too narrowly or broadly scoped, lack comprehensive privacy precautions, or not scalable to a variety of research studies [60, 129, 15, 7]. These are all challenges I also encountered while building the tools developed for the studies described in Chapters 2 to 4. In this chapter, I provide descriptions of the design and implementation details and pitfalls I encountered while developing those tools and provide recommendations for researchers or tool-developers creating their own data export donation and transfer systems.

5.2 Background and Related Work

In this section, I describe existing tools for processing and making sense of data exports (also commonly known as data donation packets, DDPs, when used for research). I then provide an overview of personal data exploration tools not necessarily aimed for research studies, but for internet users to explore their data.

Existing Data Donation Tools. Although much research has made use of data exports (see Section 4.2), only a handful of tools are open-source, actively maintained, well documented, and extendable to a variety of study types. In fact, I only identified two that meet these criteria: Data Donation Task (DDT, which overlaps and builds upon Feldspar, the paid version of the product, and OS2DF and Port, DDT’s first iterations) [15, 7] and the Data Donation Module (DDM) [129].

In addition to DDT and DDM, University of York’s Smart Data Donation Service (SDDS) seems to provide a similar service [186]. Though, it is unclear from their website exactly what features their software support, if it can be self-hosted, and the specific technology stack their project uses. It stands out from other tools, though, in that providing a usable database of donations is part of their contribution.

Other tools seem to be in early stages still, such as Princeton’s Digital Data Donation Initiative [141]. Others, like the Broad Institute’s Pepper, has the much broader scope of providing end-to-end study participant management and provides limited configurability of data export processing [23] (note the project may be stale since their documentation page has not been updated in over six years). WhatsApp Explorer, while recent and seemingly easy to set up, is built for a specific platform data source (WhatsApp) and record type (group chats) [60].

Areas for Improvement in Data Donation Tools. There are three main areas for improvement in existing data donation tools. First, all but one of the open-source tools

require knowledge of programming for more flexible uses (DDM is the exception). While this expertise may seem like a requirement for setting up a data donation system, many tools aimed at non-coders exist for building websites with custom functionality and data hosting (e.g., Squarespace¹ and WordPress²) and modular content management systems (CSMs), some of which are even open-source (e.g., Wagtail³).

The second is that tools are needlessly restrictive for the purpose of data donations for research studies in particular, despite the most complex tasks in each of their tools are the intake, parsing, processing (which often involves filtering or anonymization), and transfer of data exports. A tool that only does these data tasks that can be used in a browser or server context would be useful for researchers who have their own specialized needs and for consumers not involved in any research at all, but who are interested in offloading data export and transfers to a local library for processing.

Lastly, no tool so far has leveraged data portability or transfer APIs in their design, despite moving consumer data is the key component of the data donation task. I outline possible reasons for this in Section 5.4.

Personal Data Exploration and Usable Privacy Tools. Researchers, internet users, and companies have long been building tools and dashboards for people to interface and interact with privacy topics (e.g., their own privacy settings). As discussed in Section 2.2.1, making these interfaces clear and useful to users is challenging, especially when designed by the companies whose privacy practices are the target of such interfaces. While many tools use browser extensions to mimic the data collection done by companies’ platforms to convey the impact of tracking to the user [178, 67, 177], others have taken to data exports themselves to accomplish this task [174, 78]. While TransparencyVis, the most notable in the latter

1. <https://www.squarespace.com/>

2. <https://wordpress.com/>

3. <https://wagtail.org/>

group, indeed developed features and visualizations specifically for data exports, only one of these features is directly related to privacy (in the “Sensitive Data” tab) [146, 153, 47]. Additionally, it only supports eight platforms. This is a gap in the potential use of data exports to encourage internet users to think of their privacy online.

5.3 Data Export Web Applications and Tools

In this section, I describe the goals, design, and technical details of the different tools and web applications I developed for the studies in this thesis. The tools were designed with security, privacy, efficiency, and transparency in mind. This often led us to forego opportunities and analyses that would use more of participants’ raw data exports, since doing so would likely mean transmitting and gaining access to sensitive data.

Common Features. For every tool we deployed the transmission of participant data and technical logs occurred via HTTPS to a server managed by our research group. While our techniques for storing data changed from tool to tool, databases and directories were always access-controlled with passwords, UNIX groups, and only available to users with access to our server.

5.3.1 *Data Illuminator: Data Export Redaction and Annotation*

This web application enabled the completion of the study presented in Chapter 2. It was the first DSAR-related tool I built and provided the foundation for the development of every other tool described in this chapter.

Goals and Core Functionality

This web application was built to accomplish three goals. First, to extract all structured data to remove personally-identifiable information and to render this data in a user-friendly way,

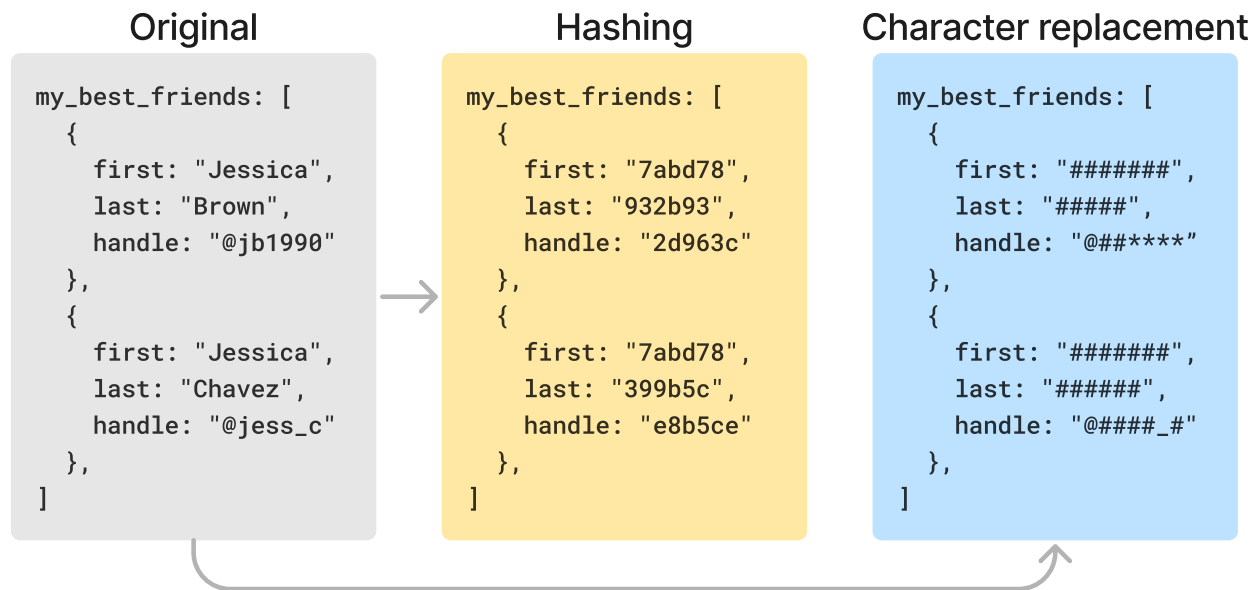
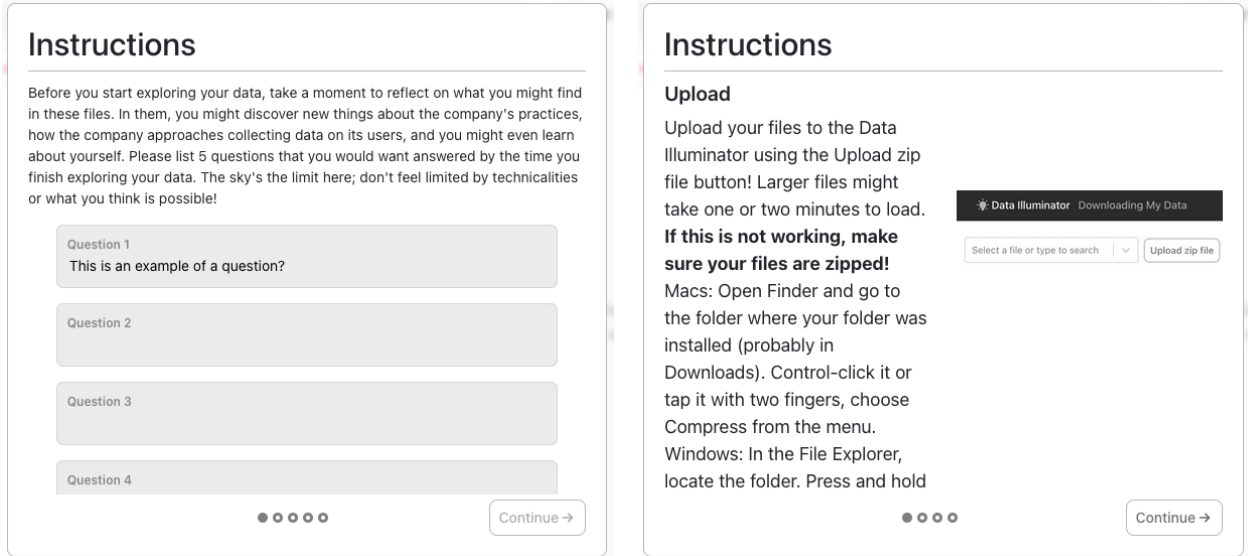


Figure 5.1: We created two anonymized versions of the data export using hashing and character replacement, respectively.

so participants could interact with it. This begins when a participant uploads a compressed file containing their data export from a platform. Data is redacted using two techniques: **hashing** and **character replacement** (see Figure 5.1). The sub-goal in this step was to maintain structural information and metadata about the export while removing all participant personally identifiable information (PII).

Our second goal was to collect participants' questions and answers to those questions. To accomplish this, we built two modal elements that the web-app renders at the beginning and end of the study. As Figure 5.2a shows, the first modal asks participants to come up with questions they would want the data export to answer. The post-study modal (Figure 5.2b) shows participants instructions for uploading and submitting their data at the end of the study and prompts them to reflect on their questions by replacing each "Question #" labels with the participant question itself.

Third, we wanted to enable participants to explore and search through their data and annotate parts that stood out to them. To enable the former, we built a fuzzy search feature for file paths and keys, column names, and values and a select element to navigate to



(a) Instructional slide prompting the participant to come up with five questions about their data. The slide post-annotation task that asks for more questions is nearly identical.

(b) Instructions for uploading their data export and completing the study.

Figure 5.2: The two modals that the Data Illuminator renders.

different files. We also paginated large files to reduce the load on participants' browsers and to make the exploration experience less overwhelming. To enable annotation, we leveraged the parsing of structured files described above to enable generating and attaching participant comments and labels to lines in those files. Figure 5.3 shows these exploration tools and the line selection for later annotation and Figure 5.4 shows the user-experience for creating and viewing annotations.

Architecture

The web-app was built between 2022 and 2023 using React 17 with Create React App tooling via react-scripts 4 and React Router v5. It uses ADM-ZIP [38], Papa Parse [69], and big-json [97] for parsing the exports and content of the exports and Dexie.js v3 [2] to interact with the browser's Indexed Database (IndexedDB) [9]. As mentioned before, no data is sent to our

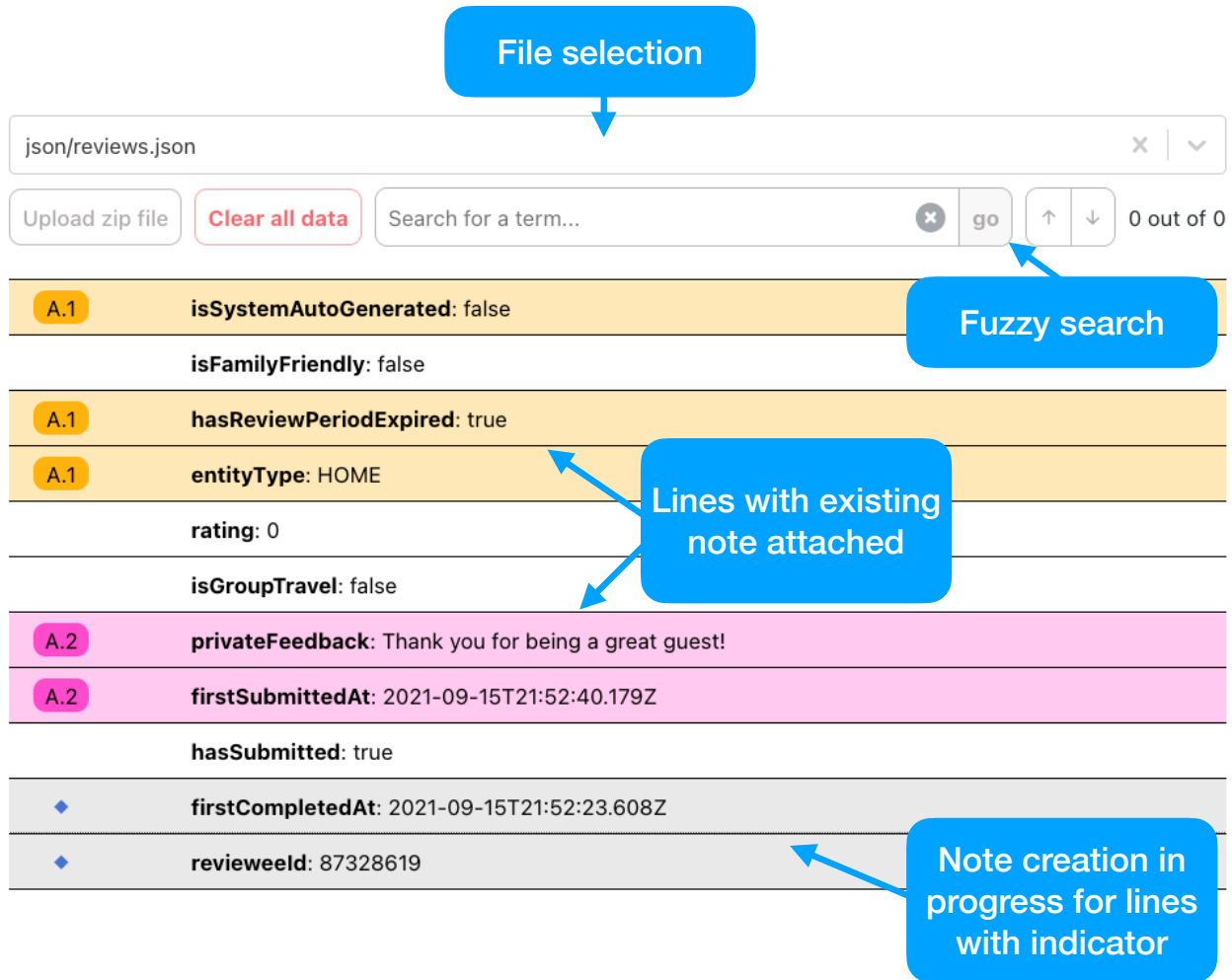
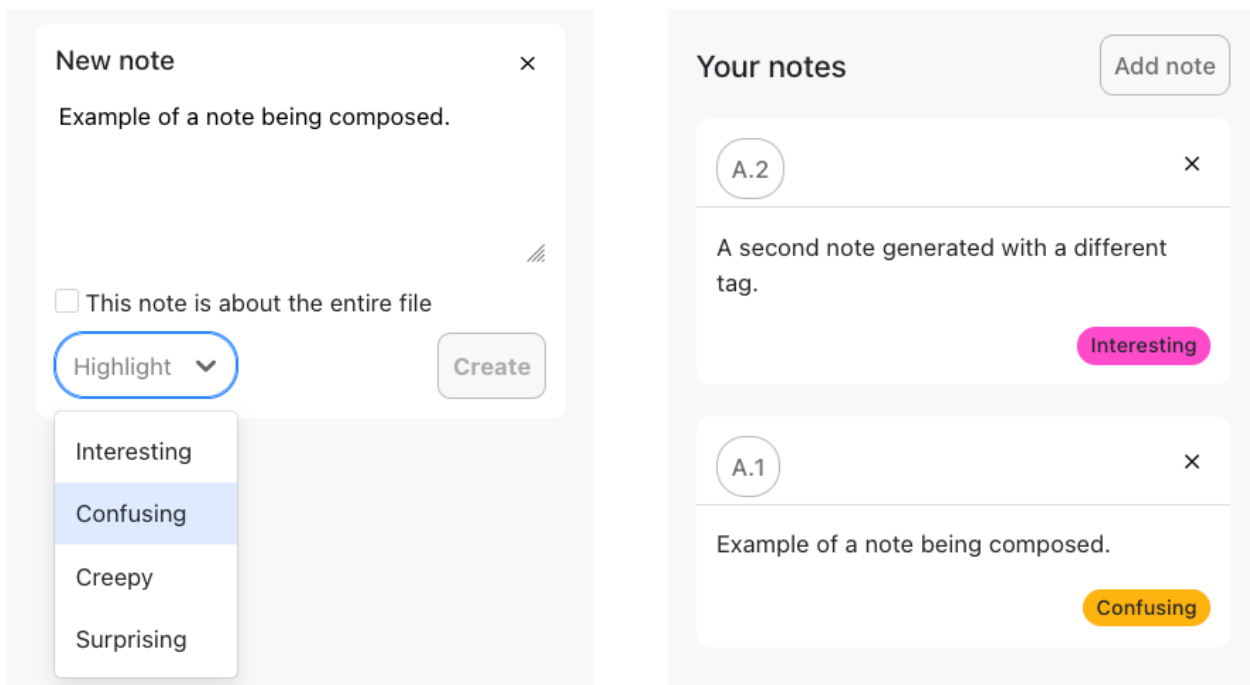


Figure 5.3: The exploration and line annotation features built into the Data Illuminator web-app.

server before the end of the study, so we chose IndexedDB as our data storage client since it supports large, unstructured files (compared to something like the browser's Local Storage), relatively quick database operations, pre-defined queries, and indexing (which we used for the fuzzy search feature). The database has separate object stores for files (FileUploads), participant notes, logs, and questions.

Once the participant uploads their compressed data export, the entire Blob is loaded into memory and each file is processed independently. First, the web-app filters out dotfiles and files of type other than JSON, CSV, and HTML and writes the remaining files into



(a) The component in the Data Illuminator web-app that allows participants to write create a comment.

(b) The existing notes created by the participant. Note identifiers (i.e., “A.1” and “A.2” correspond to highlighted lines in Figure 5.3).

Figure 5.4: The note creation and viewing feature in the Data Illuminator web-app. Each note has a “highlight,” lines in a file or a whole file associated with it, an identifier, and the participant’s text comment.

FileUploads. The web-app only converts the content of the files into a set of line objects once the participant selects that file. This happens on-demand, and the lines are not stored in their own object store (see Section 5.3.1). Only the indices of highlighted lines are written to the FileUploads in an array. The web-app also detects search matches during the parsing of the lines in a file.

The web-app handles two special parsing situations. First, it transforms the content of CSV files to JSON for rendering consistency. Second, since the web-app only supported five platforms, we decided to build Google-specific parsing for `MyActivity.html` files, since these were all structured identically and held much of the participant data contained in their export.

The two redaction steps (hashing and character replacement) only happen at the end of

the study, when the participant clicks the submit button. When hashing, the web-app first splits values by white space, returning a list of tokens. Tokens are then salted on the client side with a randomly generated 32-byte string unique to each participant, which is never sent to the server and would be computationally infeasible to brute force. This hashed version allowed us to identify repeated values since a given term with a fixed salt always hashes to the same value. The web app also creates a separate version of the data export in which values undergo character replacement. Using this technique, all letters in the values are replaced with pound characters (#) and all digits with asterisks (*). Character replacement preserved the file size and the length of values, enabling additional analyses.

On submission, the web-app collects all the data mentioned so far (except for non-redacted file content) and sends it to the server via an HTTPS POST request. The server (running Apache) reads the payload and recreates the directory structure in the participant exports in an access-restricted directory in the file system. Notes and logs are written to separate CSV files (also access-restricted).

Improvement Opportunities

We identified several bottlenecks and flaws in our design of this first web-app. One such flaw is the web-app makes no use of parallelism to process and transform data, meaning the UI freezes until this process is completed. This also means that, when the submission process begins, files are redacted sequentially, despite not depending on each other. This inefficient data handling also prevents the web-app from accepting and rendering files over 50MB. The package we chose for decompressing only accepts specific ZIP files, so tar-compressed files, single-file compressed ZIP files, nested compressed files, and password-protected ZIP files are not supported. The data model has several opportunities for improvement too, such as storing file content lines to avoid duplicate parsing and indexing fewer database fields.



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As a reminder, within one hour of our server receiving your files, we will use Microsoft's Presidio data protection and de-identification software tool to automatically remove things that look like:

- credit card
- email address
- date/time
- name
- phone number

- location
- US Driver's License
- US Bank Number
- US ITIN
- US SSN

Please enter the email you used to submit the screening survey.

Email

Please enter the company you're submitting your data for.

Company name

Please submit your data as a zip file using the button below.

Upload and submit

Figure 5.5: Frontend interface for platform work study participants to upload their data exports. In addition to a brief explanation of the study and privacy-protection steps we took, the web-app only asks for the participant’s email, platform for which they work, and data export.

5.3.2 *Convex: Smart Data Export Anonymization Backend*

We built Convex in 2024 for a study of the data exports of platform workers (i.e., gig workers) that I do not mention in previous chapters of this thesis since we did not complete it. In addition to recruiting challenges, we found that the few exports we did collect (for DoorDash, Grubhub, Lyft, and Uber) were sparse and contained only a subset of the files and fields that exist in the consumer versions of these exports. Finally, the third-party tools I discuss in Section 5.3.2 were not effective enough for us to continue the study ethically.

Goals and Core Functionality

Unlike the other studies for which we built the tools described in this chapter, one of the explicit goals of the Convex tool was to give researchers a comprehensive view of the data exports of platform workers (not just data export structure). Therefore, rather than taking the more conservative redaction techniques used by other tools, we tested software that allowed for more fine-grained PII redaction. Unlike the Data Illuminator (Section 5.3.1), Convex’s initial goals excluded showing participants a more user-friendly version of their exports. Therefore the front-end of the web-app is simple, as shown in Figure 5.5. Once the export file is loaded into the browser, it is immediately sent to a POST-endpoint, exposed on the lab’s server, alongside the fields in the form.

The server app first recreates the directory structure of the export file (excluding files that are not text-based, e.g., images and videos) in an access-controlled directory. Then, it schedules a server job to recursively traverse the export file system tree, anonymize those files, and write the results in a directory parallel to the original. Once anonymization is complete, the job script deletes the raw export directory.

Architecture

The front-end application was built in React 18 and the server application was built in Python as a Flask app [124]. The anonymization relies on Presidio, a context aware and customizable PII de-identification service for text and images developed by Microsoft [36]. In 2024, Presidio was the industry standard for open-source PII redaction tools [34]. We used a pre-trained model [10] to build out an entity recognizer for PII since the default Presidio data types were insufficient. Our app was written to recognize credit card numbers, email addresses, dates and times, names, phone numbers, locations, U.S. Driver’s License numbers, U.S. Bank numbers, U.S. taxpayer identification numbers, and U.S. social security numbers. These entities were then replaced with token identifiers using Presidio’s BatchAnonymiz-

erEngine. Unfortunately, the anonymization pipeline performed poorly. It was unable to recognize most street addresses and names since these were often less structured than something like a phone number.

Improvement Opportunities

As mentioned in the opening of this section, we discontinued use of Convex and the study for which it was developed due to the poor performance of the redaction technique we adopted. We believe its poor performance was likely due to the limited amount of contextual information in structured data files (like JSON and CSV). In prose, there are several contextual clues (e.g., grammar) that may indicate the presence of a sensitive entity. Passing parent keys (for JSON files) and CSV column names as context to the entity recognizer did not improve its performance enough for us to continue using the tool. Due to the massive improvement in performance in NLP models and LLMs more generally since 2024, future work could explore updating the tooling that Convex uses to more comprehensively de-identify data exports.

5.3.3 Data Donator: Efficient Client-Side Data Export Redaction

Goals and Core Functionality

We developed the Data Donator for the data export measurement study discussed in Chapter 4. The Data Donator improves upon and corrects many of the flaws of the Data Illuminator (Section 5.3.1). But, like Convex (Section 5.3.2), its interface, shown in Figure 5.6, does not have any features to render the content of data exports to its users. In contrast to all tools discussed so far, this tool supports password-protected, single-file, and non-ZIP compressed files. It was also built to capture comprehensive metadata and structural information from data exports, such as creation dates and names and sizes of all files (including unstructured and non-text files) and to handle multi-gigabyte files. It improves user-experience by



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UChicago Personal Data Requests Study: Submission Page

Thanks again for participating in our study! Verify your information and submit your data below.

Platform this data is from (required):

Select a platform to submit

Your email or passphrase (required):

example@mail.com

If your zip file(s) is password protected, please enter the password for the zip file (**NOT** your account password) below. We only use this to access your password-protected files before removing personally identifiable information from them.

Zip file password (if applicable):

Example: zipFile123!

If the platform sent you multiple zip files, please upload and submit one at a time with the button below.

Upload and submit data zip file

Figure 5.6: Frontend interface for study participants to upload their data exports, which are redacted client-side.

showing the participant a modal that tracks data processing progress, which is not frozen due to the parallelization described below.

Architecture

The Data Donator is a single-page web-app with client-side data processing with the same privacy requirements as the Data Illuminator (Section 5.3.1). In contrast to the Data Illuminator, though, it uses a modern tech stack (Vite, React 19, and TypeScript) with more powerful packages for reading (Zip.js [99]) and processing data exports, with support for HTML (via linkedom [63]) and JSONL parsing. It also makes extensive use of parallelism through the use of web workers.

Once a participant uploads a data export, a reference to the file data (stored in a Blob) is written to IndexedDB and a web worker is created. The worker thread reads the structural information of the compressed file and, without reading the actual content of the file, creates an entry for each file in the export along with file metadata (i.e., creation time, compression algorithm, encryption status, compressed and uncompressed sizes, and path). After capturing this metadata, the worker thread reads HTML, JSON (including JSONL, NDJSON, and geoJSON), CSV, TSV, XLSX files that are under 200 MB into memory and then immediately writes that content to IndexedDB. If these files are over 200 MB, only a Blob reference is written to IndexedDB to avoid loading too much data into memory. The worker thread discards the content of files of other types since they cannot be redacted. This metadata and file content extraction happens for each file in parallel.

After the data export file is unzipped and intra-export file metadata is stored, two additional workers are launched: one for each of the redaction techniques (hashing and character replacement described in Section 5.3.1). Large files (over 200MB) are read from a stream, redacted, and stream-written to IndexedDB. Much like the original file metadata collection, redaction also occurs in parallel for each file.

There are many other strategies to improve efficiency in and observability of the web-app. Where possible, database read and write operations are batched. Additionally, JSONL files (which, deceptively, end in “.json” like traditional JSON files) are read and redacted in batches of 10MB. Logs are sent continuously rather than as part of the final submission, allowing us to catch bugs and errors as they arise.

Improvement Opportunities

The primary design flaw of the Data Donator is the memory-heavy requirements of the redacted data export submission to the backend server. Rather than incrementally sending redacted data and metadata to the server, the web-app combines all this data into a sin-

gle payload at the end of the data processing pipeline. This aggregation requires loading all the redacted files and their corresponding metadata into memory, which can be easily overwhelmed since data exports surpass 1 GB in size.

While data processing in the web-app is fairly modular, it could be further encapsulated in its own package, which the web-app (and other web-apps) could then import as a dependency. This isolation can encourage the data processing functionality to be used for reasons other than data donation, such as personal data exploration. Finally, files written in XML could be parsed and redacted using similar techniques to JSON, CSV, and HTML files to provide a more comprehensive view of the data exports. Future work could also explore ways of more intelligently redacting TXT and PDF files to preserve some structure, since these were common in the data exports collected for the study described in Chapter 4.

5.3.4 DSAR Explorer: Privacy Provocations via Data Export Visualization

Goals and Core Functionality

Like the Data Illuminator (Section 5.3.1), the DSAR Explorer is a user-centered web-app whose main goal is to provide more accessible ways for consumers to interact with their data exports. The primary goal of the DSAR Explorer is to encourage its audience to reflect on the data practices of the platforms they use through features and visualizations built around their data exports. Unlike the other tools, data export redaction is not a goal of the web-app, though it may be in the future (see Section 5.3.4).

In this tool, each export receives its own dashboard. Figure 5.7 shows the dashboard for my Uber data export generated in 2025. The user can navigate between dashboards using a drop-down select element in the navigation bar. A **card**, for the purposes of this web-app, is a modular UI element that encapsulates some information about the data export (e.g., Figure 5.8). Currently, there are six types of cards: file count, record count, directory count, location boolean, record comparison, and location visualizer. The first four are text-based

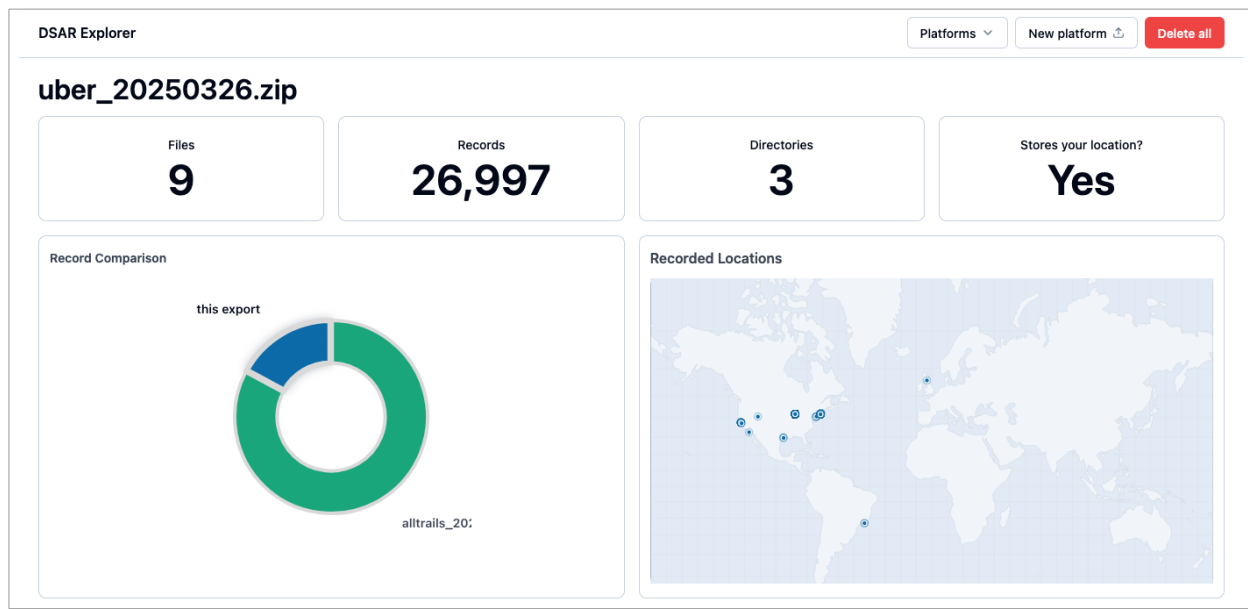


Figure 5.7: Frontend interface for the DSAR Explorer. At the top is the navigation bar, where a user can upload new exports, navigate between exports, and delete all uploaded exports. In the center is the selected export dashboard, with different cards to encourage them to think about their privacy.

and communicate static information about the data export. The purpose of these four is to encourage the user to think about the scale of data the platform has collected on them.

The remaining two cards are data synthesis cards. The record comparison card prompts the user to compare data practices between platforms they use. Doing so highlights how some platforms collect more data than others and that the scale of data collection is not necessarily correlated to the utility of the platform to the consumer. The location visualizer shows all the location records extracted from the data export on a world map. Here, the provocation not only leverages the scale of data collection, but also the subjective qualities of the data by showing the user all the locations where the platform has tracked the user.

Architecture

The design of this web-app builds on that of the DSAR Explorer by incorporating a storage mechanism for the content of all export files, not just for files that can be redacted. This

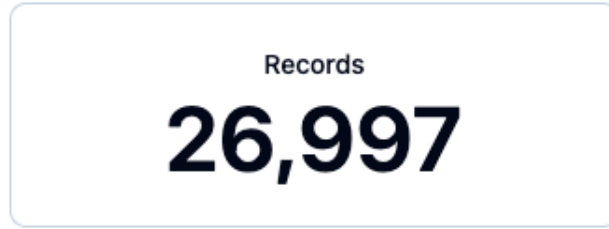


Figure 5.8: A record count card, which shows the user how many records there are in the data export selected.

requirement is due to this web-app's broader scope, which aims to show the user summaries and syntheses from their unredacted data.

While IndexedDB is one of the most efficient solutions for browser-side storage, the origin private file system (OPFS) is another a popular choice [111]. The OPFS shares the same disk space as IndexedDB, but is optimized for large (over 3,000) bulk read and write operations, in which it has on average better performance than IndexedDB [110]. The main downside to using OPFS is that it has less specialized functionality than IndexedDB since the interface it exposes is essentially the same as a traditional file system: files and directories can be created, written to, read from, and deleted. Despite the limitations, I opted to use the OPFS given that one of the primary storage tasks in this web-app is to store arbitrarily large files from the data export and occasionally read from them to perform higher level data processing.

In my hybrid approach, the content of each file is stored in an OPFS file and information about that file is stored in IndexedDB. That information includes the original file path as it appears in the export, computed measures (e.g., record count), other file metadata, and the OPFS file path. Any time a read or write operation to a file is required, the OPFS file path is first fetched from IndexedDB and then used to complete that read or write operation. This ensures that, most of the time, a minimal amount of data from the file is loaded into memory and the potentially large content of the file remains untouched. In fact, the only times a OPFS file is read from in the current version of the web-app is when the dashboard

for that file’s export is first opened. Efficiency is further achieved by relying heavily on web workers to parallelize data processing, as with the Data Donator.

Improvement Opportunities

There are three areas of investment that could improve the DSAR Explorer. First, additional platform-agnostic cards should be added to expand the potential impact of the web-app by encouraging users to reflect on their privacy in different ways. These could involve computing and comparing the number of files of different types (e.g., images) to convey to the user more qualitative information about the data stored in their exports. There is also opportunity to develop platform-specific cards given the knowledge of the types of records and data formats provided by the schemas I present in Section 4.4.2. Second, redaction tasks similar to those in the other tools in this section could be added to the web-app so that more studies investigating the make-up and content of data exports can be run. Finally, privacy actions should be encouraged by the web-app too, such as by linking to platforms’ settings pages or to information about making other types of data subject requests (e.g., deletion). By providing instructions for enacting privacy preferences or by building these steps into the web-app itself, the web-app can not only inform users, but also empower them to ensure their privacy preferences are being respected.

5.4 Pardner: A Data Transfer Tool Leveraging Portability

In this section, I describe the work I have done in collaboration with the Data Transfer Initiative⁴ on (1) enabling and facilitating access and fetching of personal data and (2) leveraging data portability, rather than access, to build data donation mechanisms for research. Both tools I discuss below are public and open-source [17, 18].

4. <https://dtinit.org/>

Ideally, data portability would be a more useful mechanism for accessing consumer data for research purposes. Unlike in the access scenario, data can be sent directly from the platform from which the researcher is seeking data to the researcher themselves, without requiring coordination with the study participant (besides authorizing the transfer), which is one of the primary barriers in conducting research that relies on data donations [182]. However, as I discuss below, there are many barriers to making data portability and platform-built portability APIs useful to most researchers.

Pardner. The first of the two tools I built is Pardner: a Python library for authorizing access and fetching personal data from portability APIs and platforms. With the tool, we aimed to facilitate, streamline, and standardize access to data associated with a user on an online platform. This is done via a common interface to execute OAuth 2.0 on a variety of platforms and access the public APIs of those platforms. The tool then exposes platform-agnostic data fetching functions for different data categories. E.g., a user looking to fetch their 10 most recent exercise activities on Strava, an exercise tracking app, can do so by calling the `fetch_physical_activity_vertical` method on the Strava service class, after setting up the proper credentials. The possible uses of Pardner are much larger than just enabling data donations, but that is the proof-of-concept we decided to build to show the power of the tool.

Pardner-site. We built Pardner-site to demonstrate a possible use of Pardner: as a facilitator for data donations for research. The entire donation flow can happen in one pass: participants can authorize fine-grained donation (i.e., allowing transfer of only user-specified data categories) once and Pardner-site, using either the platform’s public APIs or portability APIs, can fetch the desired personal data directly from the platform. The site is built using Django in such a way that it is configurable by anyone, not just software engineers or computer scientists, on the study and data category level.

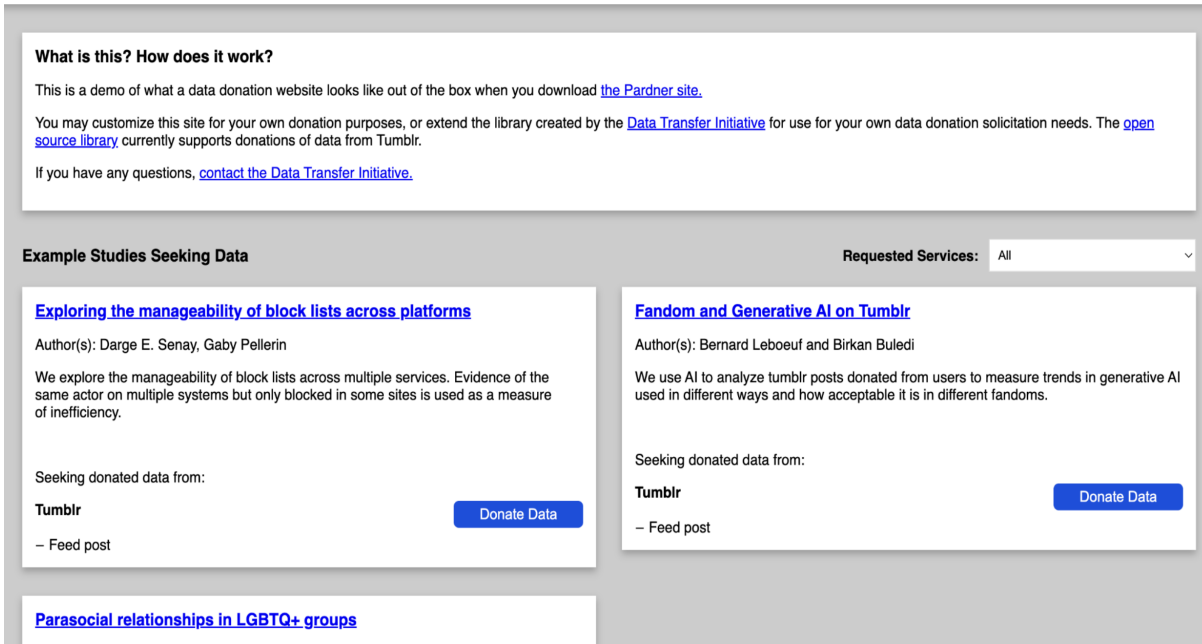


Figure 5.9: The landing page for Pardner-site, with different studies seeking different types of data that a prospective participant can choose from.

So far, we have implemented data requests for services that have OAuth/HTTP APIs and low barriers to 3rd party API access. If the site was hosted by an actual research organization able to apply to services for 3rd party API access, many kinds of data could be available, such as:

- Messages, social media posts and comments, block lists and follow lists, from social media and forum sites like Tumblr, Instagram, TikTok, and Reddit.
- Personal Internet usage history such as search history, AI conversations media views, music or podcast listens, ad impressions.
- Heart rate or blood-glucose or other health monitor data.
- Exercise or food tracking apps like Strava.

Table 5.1: Most relevant portability APIs by service and approval process classification. “Immediate” platforms auto-approve requests to their APIs, “Approval form” do so after filling out a single form, and “Extensive application” only do so after a length application involving communication with a point-of-contact at the platform.

Service	Examples of available data	Approval process
Spotify	saved media (songs, playlists, etc.); 50 recently played tracks	Immediate
Stack Exchange	questions; answers; comments; notification/inbox	Immediate
Strava	all physical activity instances (plenty of metadata); likes; comments	Immediate
Tumblr	posts on feed; user posts; comments; blocked accounts	Immediate
GroupMe	messages; chats/group chats	Immediate
Reddit	all posts per user/Subreddit; messages	Approval form
Vimeo	videos posted; comments; videos watched	Approval form
Amazon	ads clicked; orders; subscriptions; search history	Extensive application
LinkedIn	posts; messages; network	Extensive application
Uber Driver	trips	Extensive application
Google	ads settings changes; chrome extensions and settings; maps: saved commutes, posts, reviews; play store: installs; search: end-user reviews (for places or services); YouTube: all videos posted, watch list, subscriptions, comments	Extensive application
Meta (Facebook and Instagram)	messages; posts; network	Extensive application
TikTok	posts; watch history; browsing/search history; location data	Extensive application

Barriers to leveraging data portability in research. While the proof-of-concept is functional with certain platforms (i.e., Strava, Tumblr, and GroupMe), adapting it to work with others would be challenging. During the planning phase of this project, I audited the APIs of 24 social and consumer-facing platforms by conducting searches using the keywords: portability API, public API, and free API. Of these, 11 were excluded because their APIs were commercial only. From the remaining 13, eight had APIs I could apply to access, but, as Table 5.1 shows, the process of obtaining this access could take several weeks. Even for the five platforms with immediate access, all had restrictions on the number of authenticated users allowed per API key, making it impossible to deploy a tool like Pardner-site without obtaining further access. While this level of security is understandable since portability APIs can expose a vast amount of personal data from a user, it restricts the API’s use to

commercial actors and excludes researchers and individual users who seek to access personal data on the platform.

5.5 Conclusion

In this chapter, I presented six tools that provide access to consumer personal data stored by online platforms. The first four were built as part of the studies I presented in Chapters 2 to 4. These processed study participants' data exports and redacted them using different methods to preserve participant privacy. Many of these involved client-side parsing to further minimize the amount of personal data leaving the participants' machines. The last two tools I describe in the chapter were built in collaboration with the Data Transfer Initiative. These leverage the portability APIs and generic public APIs provided by large online platforms. While data portability may seem like a promising direction in enabling secure data donations with low burden to the participant, the barriers to accessing companies' APIs means this is not as feasible as leveraging data exports for data donation.

CHAPTER 6

CONCLUSION

6.1 Reflections on the State of Data Subject Access Requests

A decade after dozens of countries started requiring companies to process and respond to data subject access requests (DSAR), the state of the implementation of this right is still rapidly evolving. While some companies are converging on similar UIs and request mechanisms (e.g., due to using the same vendors), others are diverging and building highly complex and configurable interfaces (e.g., Google and Meta). After investigating the entire DSAR consumer journey, from finding out how to make a request to unzipping the data export and exploring the files within, the findings I present in this dissertation show that consumers still face many barriers when it comes to getting access to and understanding their personal data.

In Chapter 2, I presented a study on consumer reactions to making DSARs. In it, I showed that consumers have many questions about their data and the data practices of the platforms they use that cannot be answered by data exports in their current form. Despite these shortcomings, participants in the study still had positive experiences when remembering life events or experiences and learning about themselves through exploring their data. In Chapter 3, I described the three forms DSARs mechanisms and defined over a dozen measures and features that were used in the 250 platforms we analyzed. I showed that there is variability in the complexity, ease of access, configurability, authentication security, and clarity in DSAR mechanisms. Chapter 4 explored the other side of DSARs: the data exports resulting from making these requests. Much like in the request, exports varied in size, complexity, detail, and accessibility. In addition to describing exports, I also presented the Data Export Schema Repository, which has structural information about the data exports (e.g., file schemas) for the 69 platforms in that study. Finally, throughout

Chapter 5 I presented practical tools for using data exports for research, including tools for processing those data exports in privacy-preserving ways.

The future of DSARs would benefit from policymakers and product and engineering teams focusing their attention on what meaningful access to personal data looks like. The current approaches, with all their differences and complexity, are not meeting consumers' needs. Defining JSON keys and CSV column names, creating indices of data export file structures, providing syntheses and summaries of data practices, and coupling privacy-preserving actions with data exploration are all steps in the direction of enabling meaningful data access. Beyond these, though, there are still many opportunities for future research into identifying how exactly DSARs and data exports can enable consumers to have more agency over their data.

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